A construction of four-dimensional Yang–Mills theory

First sections for review; the detailed proofs are incomplete.

Contents

I The main theorem **2**

1 The result **2**

2 Definitions **2**

3 The main theorem **10**

4 Strategy **11**

5 The principal propositions **12**

II The construction **17**

6 Lattice fields **19**

7 Local polynomials and activities **26**

8 Infinite volume **34**

9 Free Maxwell fields **49**

Part I
The main theorem

1 The result

How can a model on a lattice give a quantum field theory on continuous space and time? We study this question for four-dimensional Yang–Mills theory, starting with Wilson's lattice model. On a finite lattice the variables are finitely many unitary matrices, and their law is an ordinary probability measure. The coupling is the parameter controlling the weight assigned to the lattice energy; a local observable is a function of matrices near one point. The problem is to choose them so that their correlations—the averages of their products—have a limit as the lattice becomes finer and the box becomes larger.

We construct such limits for matrix groups with simple nonabelian Lie algebra (Definition 2.1). The spacing tends to zero first, at each fixed box size. From any increasing sequence of box sizes, one can then select a subsequence along which all the required correlations converge. The limiting correlations are averages of component correlations, called phases. One selected phase gives quantum fields on a Hilbert space, with positive energy, relativistic covariance and commutativity at spacelike separation (Definition 2.16). The same local product expansions, replacing nearby fields by sums of fields at one point, hold in the average and in every phase (Definition 2.20). A logarithmic correction to the correlation of two copies of one field distinguishes it from the finite free-field expressions specified below. The result does not establish a mass gap, a positive lower bound on energies above the vacuum.

Part I begins with definitions of the lattice model and its continuum objects, followed by the precise theorem. The strategy then separates the proof into five results: constructing the finite-volume limits, passing to infinite volume and its phases, reconstructing relativistic fields, expanding local products, and proving the logarithmic two-point law. Part II develops their proofs. It begins with local lattice observables, studies what happens when short-distance variables are integrated out, and then follows the resulting correlations into the continuum theory. This order keeps the finite probability model in view while explaining how its limiting correlations become quantum fields.

The statements are matched to the fixed formal-source revision in [1].

2 Definitions

We begin with the finite-dimensional lattice model and the observables whose averages will converge. We then specify the continuum correlations, their decomposition into phases and the relativistic fields reconstructed from one phase. The last definitions give the two short-distance tests in the theorem. Throughout, $\mathbb{N} = \{0, 1, \dots\}$; empty products equal 1, and correlation functions of order zero are included.

2.1 The lattice and its measure

Definition 2.1 (Gauge group)**.** A gauge group consists of an integer $m \geq 1$, a closed subgroup G of the group $U(m)$ of unitary $m \times m$ matrices, and a real linear subspace $\mathfrak{g}$ of the complex matrix space $M_m(\mathbb{C})$ such that

$$X \in \mathfrak{g} \quad \Longleftrightarrow \quad \exp(sX) \in G \text{ for every } s \in \mathbb{R}\,. \tag{2.1}$$

Its Lie algebra $\mathfrak{g}$ is simple if $[X,Y]=XY-YX\neq 0$ for some $X,Y\in\mathfrak{g}$, and every real subspace $J\subset\mathfrak{g}$ with $[\mathfrak{g},J]\subset J$ equals 0 or $\mathfrak{g}$.

The matrices in G will be the variables of the lattice model. The specified embedding in $U(m)$ fixes the trace used in its energy. We do not assume that G is connected.

Definition 2.2 (Periodic lattice)**.** For G as in Definition 2.1 and a positive integer ℓ, let $\Lambda_\ell=(\mathbb{Z}/\ell\mathbb{Z})^4$ and write e_μ, $1\le\mu\le 4$, for its coordinate unit vectors. The sites are the points of Λ_ℓ. A positive link (y,μ) joins y to $y+e_\mu$. A link configuration is a map

$$U:\Lambda_\ell\times\{1,2,3,4\}\longrightarrow G\,. \tag{2.2}$$

The value assigned to the reversed link is $U(y,\mu)^{-1}$.

A link matrix describes parallel transport along one edge. Multiplying four of these matrices around an elementary square compares transport along its two coordinate paths. This product, the plaquette matrix, is the local quantity from which the Wilson energy is formed.

Definition 2.3 (Plaquettes and energy)**.** For a configuration as in Definition 2.2, set

$$\begin{aligned} U_{y;\mu\nu}&=U(y,\mu)U(y+e_\mu,\nu)U(y+e_\nu,\mu)^{-1}U(y,\nu)^{-1},\\ E_\ell(U)&=\sum_{y\in\Lambda_\ell}\sum_{\mu<\nu}\left(1-\frac{1}{m}\operatorname{Re}\operatorname{tr}U_{y;\mu\nu}\right). \end{aligned} \tag{2.3}$$

The energy counts each unordered pair of coordinate directions once. Each summand lies in $[0,2]$, so $0\le E_\ell\le 12\ell^4$. The probability measure weights configurations according to this energy.

Definition 2.4 (Wilson measure)**.** Let G be as in Definition 2.1 and let h be normalised Haar measure on G, the left-translation-invariant probability measure. For $\beta\in\mathbb{R}$, the Wilson measure on the configurations of Definition 2.2 is

$$\begin{aligned} Z_{\ell,\beta}&=\int e^{-\beta E_\ell(U)}\prod_{y,\mu}h(dU(y,\mu)),\\ \mu_{\ell,\beta}(dU)&=\frac{e^{-\beta E_\ell(U)}}{Z_{\ell,\beta}}\prod_{y,\mu}h(dU(y,\mu))\,. \end{aligned} \tag{2.4}$$

The bound on E_ℓ makes $Z_{\ell,\beta}$ finite and strictly positive for every real β. Increasing β gives greater relative weight to configurations with smaller energy. This parameter is called the bare inverse coupling: it is specified in the original lattice measure, before any variables are integrated out. Changing coordinates in the internal matrix space independently at each site acts as follows.

Definition 2.5 (Gauge transformations)**.** For configurations as in Definition 2.2 and $\gamma:\Lambda_\ell\to G$, define

$$U^\gamma(y,\mu)=\gamma(y)U(y,\mu)\gamma(y+e_\mu)^{-1}\,. \tag{2.5}$$

An observable F on the configurations of Definition 2.2 is gauge invariant if $F(U^\gamma)=F(U)$ for every U and γ.

2.2 Local observables and the continuum limit

An observable is a function of the link matrices; its expectation is a measurement in the finite lattice model. To pass to a field at a point, we use observables that depend on links near one site. Their coefficients may change as the lattice is refined. To follow one observable through these changes, we give it a fixed index, separate from its formula on a particular lattice.

Definition 2.6 (Observable indices and insertions)**.** Let $\mathcal{T}$ be a set of observable indices. An insertion with index $v \in \mathcal{T}$ at a site $y \in \Lambda_\ell$ is a function on link configurations (Definition 2.2):

$$I(v, y; \cdot) : G^{\Lambda_\ell \times \{1,2,3,4\}} \longrightarrow \mathbb{C}, \qquad U \longmapsto I(v, y; U)\,. \tag{2.6}$$

For a finite set $\Sigma \subset \mathbb{Z}^4$, write $\Sigma \bmod \ell$ for its image in Λ_ℓ. The insertion is supported on the links inside $y + (\Sigma \bmod \ell)$ if it takes the same value on any two configurations that agree on all links whose two endpoints belong to this set.

The index set is arbitrary at this stage; no linear structure or degree is required. Our construction uses pairs consisting of a degree and a gauge-invariant polynomial in the Taylor coefficients of a classical connection, a smooth $\mathfrak{g}$-valued one-form describing parallel transport (Definition 6.4). That polynomial specifies the classical observable to be represented, while the insertion is its actual function of link matrices. The continuum field obtained from an observable index will be part of the hierarchy below. We next choose the physical sizes of the lattices. The index N refines the spacing and p increases the volume.

Definition 2.7 (Wilson family)**.** For a gauge group (Definition 2.1), a Wilson family $\mathcal{W}$ consists of strictly increasing positive integer sequences $(D_N)_{N\in\mathbb{N}}$ and $(L_p)_{p\in\mathbb{N}}$, real numbers $\beta_{N,p}$, normalised Haar measure h on G, a set $\mathcal{T}$ of observable indices, and insertions $I_{N,p}(v, y; U)$ as in Definition 2.6 on $\Lambda_{L_pD_N}$. We write

$$a_N = \frac{1}{D_N}, \qquad \ell_{N,p} = L_pD_N, \qquad \mu_{N,p} = \mu_{\ell_{N,p},\beta_{N,p}}\,. \tag{2.7}$$

The lattice has L_pD_N sites per side, separated by physical distance a_N; thus its physical side is L_p. Taking $N \to \infty$ at fixed p refines the lattice without enlarging the box. Taking $p \to \infty$ enlarges the box. The spacing a_N is a short-distance cutoff; its limit to zero is called the ultraviolet limit. The family also specifies the observables whose averages are to have limits.

Definition 2.8 (Regularity)**.** A Wilson family (Definition 2.7) is regular if:

Reg1. There exists $b_* > 0$ such that, for every $\varepsilon > 0$, there exists N_0 with

$$\left| \frac{\beta_{N,p}}{\log\left(\frac{1}{a_N}\right)} - b_* \right| < \varepsilon \qquad (N \geq N_0,\ p \in \mathbb{N})\,. \tag{2.8}$$

Reg2. Each $U \mapsto I_{N,p}(v, y; U)$ is continuous.

Reg3. Each insertion is gauge invariant in Definition 2.5.

Reg4. For every $v \in \mathcal{T}$, there exists a finite set $\Sigma_v \subset \mathbb{Z}^4$ such that every $I_{N,p}(v, y; \cdot)$ is supported on the links inside $y + (\Sigma_v \bmod \ell_{N,p})$ in Definition 2.6.

The same finite set in Reg4 works at every spacing, volume and site. Its diameter in physical units tends to zero with a_N. Condition Reg1 is a logarithmic growth law for the bare inverse coupling, uniform in the volume; its positive slope is left unspecified here.

Definition 2.9 (Physical positions and moments)**.** Let $r_\ell(j)$ be the representative of $j \in \mathbb{Z}/\ell\mathbb{Z}$ in $\left(-\frac{\ell}{2}, \frac{\ell}{2}\right]$. The physical position of $y \in \Lambda_{\ell_{N,p}}$ is

$$\hat{y} = a_N \big(r_{\ell_{N,p}}(y^\mu)\big)_{\mu=1}^{4} . \tag{2.9}$$

For a Wilson family (Definition 2.7) and $v = (v_1, \dots, v_n) \in \mathcal{T}^n$, define the moment and its smearing by a function f by

$$\begin{aligned} M_{N,p}(v; y) &= \int \prod_{i=1}^{n} I_{N,p}(v_i, y_i; U)\, \mu_{N,p}(dU), \\ C_{N,p}(v; f) &= a_N^{4n} \sum_{y \in \Lambda^n_{\ell_{N,p}}} f(\hat{y}_1, \dots, \hat{y}_n) M_{N,p}(v; y) . \end{aligned} \tag{2.10}$$

A correlation is an average of a product of observables. Smearing averages its spatial arguments against a test function, so the lattice sum can be compared with a continuum integral. We test only configurations of distinct points.

Definition 2.10 (Separated tests)**.** For $n \in \mathbb{N}$, put

$$\mathrm{Conf}_n = \{x \in (\mathbb{R}^4)^n : x_i \neq x_j \text{ whenever } i \neq j\} . \tag{2.11}$$

A separated test is a smooth function $f : (\mathbb{R}^4)^n \to \mathbb{C}$ whose closed support is compact and contained in Conf_n. We write $\operatorname{supp} f$ for the closed support, the closure of the set on which f is nonzero.

Such a test vanishes near every coincidence of two points. It can depend jointly on all n points and need not be a product of one-point tests. To compare a torus with Euclidean space, its support must also stay inside the physical box.

Definition 2.11 (Tests supported inside a volume)**.** A separated test f (Definition 2.10) is supported inside volume p if

$$\operatorname{supp} f \subset \left(\left(-\frac{L_p}{2}, \frac{L_p}{2} \right)^4 \right)^n . \tag{2.12}$$

Definition 2.12 (Ordered limit)**.** For a Wilson family (Definition 2.7), $P : \mathbb{N} \to \mathbb{N}$ and kernels $K_n(v; x)$, convergence along P means the following. For every $n \in \mathbb{N}$, $v \in \mathcal{T}^n$ and separated test f (Definition 2.10), the test is supported inside volume $P(k)$ eventually in k, and there exist $R_p \in \mathbb{C}$ such that

$$\begin{aligned} \lim_{N \to \infty} C_{N,p}(v; f) &= R_p \qquad\qquad (f \text{ supported in volume } p), \\ \lim_{k \to \infty} R_{P(k)} &= \int_{(\mathbb{R}^4)^n} f(x) K_n(v; x)\, dx . \end{aligned} \tag{2.13}$$

Here the moments are those of Definition 2.9 and the support condition is Definition 2.11. The first limit uses the entire spacing sequence at every volume containing the test. The second limit enlarges the volume along P. At volumes not containing the support, R_p is unconstrained. A strictly increasing P contains each fixed compact test support eventually, since $L_{P(k)} \to \infty$.

Definition 2.13 (Periodic separation)**.** For a Wilson family (Definition 2.7) and volume p, the periodically separated configurations are

$$\{x \in (\mathbb{R}^4)^n : x_i - x_j \notin L_p\mathbb{Z}^4 \text{ whenever } i \neq j\}\,. \tag{2.14}$$

Distinct points can represent the same point of a torus if their difference is a period. This definition excludes those coincidences as well, and is used for fixed-volume correlation kernels.

2.3 The continuum objects

Definition 2.14 (Spacetime and Schwartz tests)**.** Write $x = (\vec{x}, t(x)) \in \mathbb{R}^3 \times \mathbb{R}$ for spacetime, with $t(x) = x^4$, Euclidean norm $|x|$ and

$$\mathsf{m}(x) = t(x)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{x : |\vec{x}| \leq t(x)\}\,. \tag{2.15}$$

A vector is spacelike if $\mathsf{m}(x) < 0$. Let $\mathcal{S} = \mathcal{S}(\mathbb{R}^4; \mathbb{C})$, the smooth complex functions whose derivatives decay faster than every inverse power of $|x|$, with the corresponding Schwartz topology.

Hilbert inner products are antilinear in the first variable. The next definition lists the Hilbert space and fields before imposing their physical properties. A field is smeared against a Schwartz function because its value at a single point need not be an operator.

Definition 2.15 (Relativistic data)**.** A relativistic datum consists of a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ of field labels, a linear subspace $\mathcal{D} \subset \mathcal{H}$, a vector $\Omega \in \mathcal{D}$ called the vacuum, and fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$ linear over $\mathbb{C}$ in $\ell \in \mathcal{L}$ and $f \in \mathcal{S}$ (Definition 2.14). The field label ℓ specifies which quantum field is smeared; the test function f specifies how it is tested in space and time. It includes an antilinear map $\ell \mapsto \ell^*$, a strongly continuous unitary translation representation $T(b)$ and a projection-valued Borel measure E on $\mathbb{R}^4$. Its diagonal finite measures satisfy $\mu_\psi(A) = \langle \psi, E(A)\psi \rangle$ on Borel sets.

There is also a strongly continuous unitary representation U of a topological group $\mathcal{P}$, identified by a homeomorphism κ with the proper orthochronous Poincaré group. Writing $\kappa(g) = (\Lambda_g, b_g)$, this identification respects identity and the multiplication

$$(\Lambda, b)(\Lambda', b') = (\Lambda\Lambda', \Lambda b' + b)\,. \tag{2.16}$$

Here Λ is invertible, has determinant 1, preserves m and maps $\overline{V}_+$ onto itself. Pure translations $\tau(b) = \kappa^{-1}(1, b)$ satisfy $U(\tau(b)) = T(b)$. For each $g \in \mathcal{P}$ there is a complex linear bijection $\rho(g)$ of $\mathcal{L}$. The action on tests is $(g \cdot f)(x) = f(\Lambda_g^{-1}(x - b_g))$.

A projection-valued Borel measure here is normalised by $E(\mathbb{R}^4) = 1$, has $E(\varnothing) = 0$, satisfies $E(A)E(B) = E(A \cap B)$ and $E(A)^* = E(A)$ on Borel sets, and is countably additive in the strong operator topology. One can assign arbitrary operators to non-Borel sets, which none of the assertions uses. The map * on field labels is not required to be an involution. No continuity or finite-dimensionality condition on ρ is included in the data.

Definition 2.16 (Wightman conditions)**.** The relativistic datum of Definition 2.15 satisfies the following conditions, with spacetime and test functions as in Definition 2.14 and $p \cdot b$ the Euclidean pairing.

W0. For every $\psi \in \mathcal{H}$ and $b \in \mathbb{R}^4$,

$$\langle \psi, T(b)\psi \rangle = \int e^{ip\cdot b}\, \mu_\psi(dp), \qquad E(\mathbb{R}^4 \setminus \overline{V}_+) = 0\,. \tag{2.17}$$

Every vector fixed by all $T(b)$ belongs to $\mathbb{C}\Omega$.

W1. The vacuum has norm one, $\mathcal{D}$ is dense, and

$$\mathcal{D} = \operatorname{span}_{\mathbb{C}}\{\Phi(\ell_1, f_1)\cdots\Phi(\ell_j, f_j)\Omega : j \in \mathbb{N},\ \ell_i \in \mathcal{L},\ f_i \in \mathcal{S}\}\,. \tag{2.18}$$

For every $\ell \in \mathcal{L}$ and $\psi \in \mathcal{D}$, the map $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ into $\mathcal{H}$ in norm. For every ℓ, f and $\psi, \chi \in \mathcal{D}$,

$$\langle \Phi(\ell, f)\psi, \chi\rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi\rangle\,. \tag{2.19}$$

W2. The maps ρ form a representation: $\rho(1) = 1$ and $\rho(gg') = \rho(g)\rho(g')$. For every $g \in \mathcal{P}$, $U(g)\mathcal{D} \subset \mathcal{D}$ and $U(g)\Omega = \Omega$. For every g, ℓ, f and $\psi \in \mathcal{D}$,

$$U(g)\Phi(\ell, f)\psi = \Phi(\rho(g)\ell, g\cdot f)U(g)\psi\,. \tag{2.20}$$

W3. If $\mathsf{m}(x-y) < 0$ for every $x \in \operatorname{supp} f$ and $y \in \operatorname{supp} f'$, then, for every $\ell, \ell' \in \mathcal{L}$ and $\psi \in \mathcal{D}$,

$$\Phi(\ell, f)\Phi(\ell', f')\psi = \Phi(\ell', f')\Phi(\ell, f)\psi\,. \tag{2.21}$$

The finite products of smeared fields in (2.18) are called field words. Their values on the vacuum span the domain itself; the empty word gives the vacuum. The symmetry group is the Poincaré group itself rather than its double cover. The sign convention in (2.17) is converted to the usual Minkowski pairing by reversing spatial momentum; the forward cone is unchanged.

We collect the continuum objects in a hierarchy: correlations, data for their decomposition into phases, relativistic fields, and coefficients for local product expansions.

Definition 2.17 (Hierarchy)**.** A hierarchy over the Wilson family of Definition 2.7 consists of the following data.

H1. A relativistic datum of Definition 2.15.

H2. For every $d \in \mathbb{N}$, a finite-dimensional complex normed space F_d and a complex linear map $\iota_d : F_d \to \mathcal{L}$. Set $\mathbb{F} = \bigsqcup_{d\in\mathbb{N}} F_d$, $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$.

H3. A map $\pi : \mathcal{T} \to \mathbb{F}$, assigning a continuum field to each observable index (Definition 2.6).

H4. A compact metric space X, called the phase space, with a Borel measure ν and a selected point $\xi_* \in X$.

H5. For every $n \in \mathbb{N}$ and $s \in X \sqcup \{\mathrm{mix}\}$, a correlation function $S^s(a; x) \in \mathbb{C}$ on $\mathbb{F}^n \times (\mathbb{R}^4)^n$. The extra symbol mix distinguishes the average state supplied by the lattice limit from the phases $\xi \in X$.

H6. For every $n \in \mathbb{N}$, a continuation $\mathsf{F}_n(a; z, \vec{x}) \in \mathbb{C}$ on $\mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n$.

H7. For every $j \in \mathbb{N}$, a finite index set Ch_j and fields $O_{j,\kappa} \in F_j$. For every finite index set I, $a \in \mathbb{F}^I$ and $\kappa \in \mathrm{Ch}_j$, there is a coefficient function $c_{j,\kappa}(a; x, o)$ on $(\mathbb{R}^4)^I \times \mathbb{R}^4$.

H8. An element $\Theta \in F_4$.

The maps in H2–H3 relate the observable indices, continuum fields and quantum field labels:

$$v \in \mathcal{T} \overset{\pi}{\mapsto} (d, A) \in \mathbb{F} \overset{\iota}{\mapsto} \ell = \iota_d A \in \mathcal{L}\,. \tag{2.22}$$

The field label specifies the operator family $f \mapsto \Phi(\ell, f)$ (Definition 2.15). In our construction, the polynomial describes the classical observable and its index follows the lattice representative.

The degree is an index used in the product estimates below. The definition imposes no inclusions between different F_d, no prescribed curvature interpretation and no compatibility of the ι_d across degrees. The functions S^s are defined even where points coincide, but the conclusions constrain them only at separated points.

2.4 Reconstruction and phases

The mixture will be the limit of lattice moments. The point ξ_* instead selects the Euclidean correlations that continue to the relativistic fields. These two roles are related by a phase average, not by identifying the two states.

Definition 2.18 (Reconstruction properties)**.** A hierarchy of Definition 2.17 has the reconstruction properties if the following hold. Here $s \in X \sqcup \{\mathrm{mix}\}$, $n \in \mathbb{N}$ and $a \in \mathbb{F}^n$.

R1. Its relativistic datum satisfies W0–W3.

R2. For every nonzero $\ell \in \mathcal{L}$, there exists $f \in \mathcal{S}$ with $\Phi(\ell, f) \neq 0$. This is faithfulness: no nonzero field label represents the zero field.

R3. Each ι_d is injective.

R4. Some $a \in \mathbb{F}$ and $f \in \mathcal{S}$ satisfy $\Phi(\iota(a), f) \neq 0$.

R5. The map $a \mapsto \iota(a)$ from $\mathbb{F}$ onto $\mathcal{L}$ is surjective.

R6. The map $\pi : \mathcal{T} \to \mathbb{F}$ is surjective.

R7. The measure ν has total mass one.

R8. Every nonempty open subset of X has positive ν-measure; this is full support.

R9. For every state s, order n, list $a \in \mathbb{F}^n$, $x \in \mathrm{Conf}_n$ and permutation σ,

$$S^s((a_{\sigma(i)})_i; (x_{\sigma(i)})_i) = S^s(a; x)\,. \tag{2.23}$$

R10. For every s, n, a, the function $S^s(a; \cdot)$ is smooth on Conf_n.

R11. For every n, a and $x \in \mathrm{Conf}_n$, the function $\xi \mapsto S^\xi(a; x)$ is continuous.

R12. For every n, a and $x \in \mathrm{Conf}_n$,

$$\int_X S^\xi(a; x)\, \nu(d\xi) = S^{\mathrm{mix}}(a; x)\,. \tag{2.24}$$

R13. For every n, a and fixed real spatial configuration $\vec{x}$, the function $z \mapsto \mathsf{F}_n(a; z, \vec{x})$ is holomorphic on

$$\mathcal{Z}_n = \{z \in \mathbb{C}^n : \mathrm{Im}\, z_1 < \cdots < \mathrm{Im}\, z_n\}\,. \tag{2.25}$$

R14. For every n, a and real configuration x with $t(x_1) < \cdots < t(x_n)$,

$$\mathsf{F}_n(a; (it(x_i))_i, (\vec{x}_i)_i) = S^{\xi_*}(a; x)\,. \tag{2.26}$$

R15. For every n, a, real heights $h_1 < \cdots < h_n$ and tests $f_1, \ldots, f_n \in \mathcal{S}$,

$$\lim_{r \downarrow 0} \int_{(\mathbb{R}^4)^n} \mathsf{F}_n(a; (t(x_i) + irh_i)_i, (\vec{x}_i)_i) \prod_{i=1}^n f_i(x_i)\, dx = \langle \Omega, \Phi(\iota(a_1), f_1) \cdots \Phi(\iota(a_n), f_n)\Omega \rangle\,. \tag{2.27}$$

The scalar on the right of (2.27) is the vacuum expectation of the indicated product of smeared fields. In (2.27) we retain the formal source's convention that the integral of a nonintegrable function is zero. No separate positive-height integrability assertion is part of R15. The integrals in the lattice limit and (2.24) are ordinary absolutely convergent integrals, by compact support and smoothness in the first case and by phase continuity and compactness in the second.

2.5 Products and the trace test

The product expansion replaces a cluster by finitely many single fields at its centre. Fields whose positions remain away from the cluster are called spectators. One choice of coefficients must work with every list of such fields and in every state. The threshold on the cluster radius, however, may depend on the fixed spectator fields.

Definition 2.19 (Cluster geometry)**.** For a hierarchy (Definition 2.17) and finite index sets I, J, write $S^s(a, x; b, y)$ for the correlation on the concatenated lists, enumerated in one fixed order on $I \sqcup J$. For $r > 0$, $M \geq 0$ and $\eta > 0$, call (x, o) an (r, M, η)-cluster if

$$|x_i - o| \leq Mr \quad (i \in I), \qquad |x_i - x_{i'}| \geq \eta r \quad (i \neq i')\,. \tag{2.28}$$

For $c > 0$, call y c-clear of o if

$$|o - y_j| \geq c \quad (j \in J), \qquad |y_j - y_{j'}| \geq c \quad (j \neq j')\,. \tag{2.29}$$

Set $w(a) = \sum_{i \in I} \deg a_i$.

Definition 2.20 (Operator product expansion)**.** For a hierarchy (Definition 2.17), the OPE property consists of the following estimates on clusters and distant points as in Definition 2.19.

O1. For every finite I, $a \in \mathbb{F}^I$, $M \geq 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\kappa \in \mathrm{Ch}_j$, there exists $\delta > 0$ such that, for every $0 < r < \delta$ and every (r, M, η)-cluster,

$$|c_{j,\kappa}(a; x, o)| \leq r^{\,j - w(a) - \varepsilon}\,. \tag{2.30}$$

O2. For every finite I, J, $a \in \mathbb{F}^I$, $b \in \mathbb{F}^J$, $M \geq 0$, $\eta, c, \varepsilon > 0$ and $D \in \mathbb{N}$, there exists $\delta > 0$ such that, for every $0 < r < \delta$, every (r, M, η)-cluster, every c-clear spectator configuration and every $s \in X \sqcup \{\mathrm{mix}\}$,

$$\left| S^s(a, x; b, y) - \sum_{j=0}^{D} \sum_{\kappa \in \mathrm{Ch}_j} c_{j,\kappa}(a; x, o)\, S^s\big((j, O_{j,\kappa}), o; b, y\big) \right| \leq r^{\,D+1-w(a)-\varepsilon}\,. \tag{2.31}$$

The coefficient functions are those already chosen in H7. They do not depend on spectators, state or truncation degree. Both estimates have constant 1, with the dependence absorbed into δ. For small enough r, $Mr < c$ ensures that all points in the tested correlations are separated. These are finite-truncation estimates; they assert neither an infinite convergent expansion nor uniqueness of its coefficients.

Definition 2.21 (The trace two-point function)**.** For the field $\Theta \in F_4$ of Definition 2.17 and $r \in \mathbb{R}$, set

$$G_\Theta(r) = S^{\xi_*}\Big((4, \Theta), (4, \Theta); -\frac{r}{2} e_4, \frac{r}{2} e_4\Big)\,. \tag{2.32}$$

Free-field two-point functions are built from the energy $E_{\mathsf{m}}(p)$ of a particle with mass m and spatial momentum p. The following class allows finitely many masses and derivatives, and finite sums of products of the resulting kernels. We use these kernels only at positive separations $r > 0$. It specifies exactly the comparisons excluded by the theorem.

Definition 2.22 (Finite Wick profiles)**.** A mode consists of $\mathsf{m} \geq 0$, integers $j, q \in \mathbb{N}$ and a continuous real function h on $\mathbb{R}^3$ satisfying $h(\lambda p) = \lambda^j h(p)$ for every $\lambda > 0$. Set

$$E_{\mathsf{m}}(p) = \sqrt{\mathsf{m}^2 + |p|^2}, \qquad K_{\mathsf{m},q,h}(r) = \int_{\mathbb{R}^3} \frac{E_{\mathsf{m}}(p)^q h(p)}{2E_{\mathsf{m}}(p)} e^{-rE_{\mathsf{m}}(p)}\, dp\,. \tag{2.33}$$

The integrand can be assigned value zero at a zero denominator, a set of measure zero. A finite Wick profile is

$$W(r) = \sum_{\alpha=1}^{A} c_\alpha \prod_{\beta=1}^{n_\alpha} K_{\mathsf{m}_{\alpha\beta},q_{\alpha\beta},h_{\alpha\beta}}(r), \qquad c_\alpha \in \mathbb{C}, \quad A, n_\alpha \in \mathbb{N}\,. \tag{2.34}$$

Empty products are allowed.

Definition 2.23 (Trace test)**.** The hierarchy of Definition 2.17 passes the trace test if the following hold, with G_Θ as in Definition 2.21 and Wick profiles as in Definition 2.22.

T1. For every $f \in \mathcal{S}$, $\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0$.

T2. The number $A_G = \frac{3\dim_{\mathbb{R}} \mathfrak{g}}{\pi^4}$ is positive.

T3. The following limit holds in $\mathbb{C}$ over all positive radii:

$$\lim_{r\downarrow 0} r^8 \left(\log \frac{1}{r}\right)^2 G_\Theta(r) = \frac{3\dim_{\mathbb{R}} \mathfrak{g}}{\pi^4}\,. \tag{2.35}$$

T4. For every finite Wick profile W and every $\delta > 0$, there exists $r \in (0, \delta)$ with $G_\Theta(r) \neq W(r)$.

The public trace test concerns the chosen degree-four field and the time axis. Its definition alone supplies no stress tensor. In the construction, a stress tensor is a tensor of fields whose integral identities express translation and rotation symmetry; its Euclidean trace is the sum of its diagonal components. The construction identifies this field with a centred stress trace, as used in the proof of Proposition 5.5 below.

3 The main theorem

The theorem ties the continuum correlations to the original Wilson measures. The lattice family is chosen once; the later choice of a volume subsequence determines one continuum hierarchy with all four conclusions below.

Theorem A. *For every gauge group $(m, G, \mathfrak{g})$ with $\mathfrak{g}$ simple (Definition 2.1), there exists a Wilson family $\mathcal{W}$ (Definition 2.7) satisfying the regularity conditions Reg1–Reg4 of Definition 2.8 such that, for every strictly increasing $p : \mathbb{N} \to \mathbb{N}$, there exist a strictly increasing $q : \mathbb{N} \to \mathbb{N}$ and a hierarchy over $\mathcal{W}$ (Definition 2.17) with the following properties.*

P1. *For every $n \in \mathbb{N}$, $v \in \mathcal{T}^n$ and separated test f (Definition 2.10), the test is supported inside volume $p(q(k))$ eventually in k (Definition 2.11), and there exist $R_j \in \mathbb{C}$ such that*

$$\begin{aligned} \lim_{N\to\infty} C_{N,j}(v; f) &= R_j && (f \text{ supported in volume } j),\\ \lim_{k\to\infty} R_{p(q(k))} &= \int f(x) S^{\mathrm{mix}}(\pi(v_1), \dots, \pi(v_n); x)\, dx\,. \end{aligned} \tag{3.1}$$

Here $C_{N,j}$ is the Wilson moment of Definition 2.9; these are the ordered limits of Definition 2.12.

P2. *The selected relativistic datum satisfies the Wightman conditions W0–W3 of Definition 2.16. The hierarchy has all reconstruction properties R1–R15 of Definition 2.18, including the selected phase and its continuation. The mixed correlations are the phase average*

$$S^{\mathrm{mix}}(a;x) = \int_X S^{\xi}(a;x)\,\nu(d\xi) \qquad (x \in \mathrm{Conf}_n)\,.$$

P3. *The single coefficient system in H7 satisfies both OPE estimates of Definition 2.20.*

P4. *The field $\Theta \in F_4$ is centred and its selected-phase two-point function (Definition 2.21) satisfies*

$$\lim_{r\downarrow 0} r^8 \left(\log\frac{1}{r}\right)^2 G_\Theta(r) = \frac{3\dim_{\mathbb{R}}\mathfrak{g}}{\pi^4} > 0\,.$$

For every finite Wick profile W (Definition 2.22) and every $\delta > 0$, there exists $r \in (0,\delta)$ with $G_\Theta(r) \neq W(r)$. These are the trace conditions T1–T4 of Definition 2.23.

The lattice data are chosen before p. The refinement, field spaces, phases, selected relativistic theory, product coefficients and trace field may depend on p. The same refinement works for all orders of correlation, observable indices and separated tests. The lattice limit in P1 is the mixture; the relativistic theory in P2 is attached to ξ_*.

The theorem does not assert uniqueness of the infinite-volume limit, independence of the limit from the lattice choices, or convergence along simultaneous cutoff and volume limits. It does not assert a mass gap, factorisation of the mixture's correlations at large separations, agreement of phases, or that an individual phase is a lattice-state limit. It gives no distributional extension across coincident points, and excludes only the finite profiles of Definition 2.22, not all composites of Gaussian fields or continuous-mass generalised free fields (Gaussian fields whose covariances involve a continuous distribution of masses). No Euclidean covariance or reflection-positivity assertion is included among P1–P4; those properties are established internally in the construction.

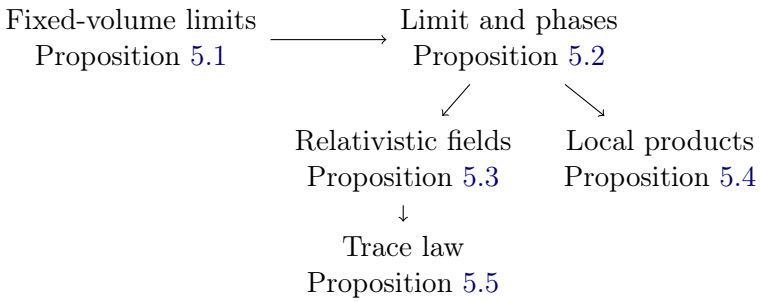
4 Strategy

The proof has two limiting operations and then three ways of reading the same limiting correlations. First we remove the lattice spacing while keeping the physical box fixed. Then we enlarge the box, allowing a subsequence of volumes. The later arguments recover relativistic fields, compare products at nearby points, and distinguish a two-point function from the specified free-field profiles. None of them may replace the correlations obtained in the first two steps.

From small spacings to a fixed box. Refining the lattice adds new fluctuations at shorter distances. We integrate these fluctuations a few scales at a time. The resulting probability law is the effective law at the larger spacing; its coupling must be compared at one fixed reference spacing. We adjust the bare coupling so that this reference coupling is always the same. The observables must be adjusted along with the probability law; otherwise their correlations need not remain finite. Proposition 5.1 produces one family of observables with these adjustments and proves their limits at each fixed volume. Its estimates are local, so they can later control regions inside arbitrarily large boxes.

From boxes to phases. Local bounds independent of box size allow a subsequence on which the correlations converge. This is the compactness step in Proposition 5.2. The result can still be an average of different states. Reflection positivity turns its positive-time correlations into inner products, and averages at widely separated times distinguish its components. After continuous rotations have been established, these averages commute and can be represented as functions on a compact space of phases. Their expectation is integration against a probability measure, giving the mixture formula (2.24). The phase construction is therefore an additional argument after compactness.

Three conclusions from one hierarchy. Proposition 5.3 selects a phase with a unique vacuum and continues its time variables to obtain relativistic fields. Proposition 5.4 replaces a product at nearby points by a finite sum of fields at one point, with a controlled error inside any correlation with distant fields. Its coefficients are chosen once, so the expansion also holds in every phase. Proposition 5.5 studies the centred trace of the stress tensor: a logarithmic correction to its two-point singularity rules out the finite Wick profiles. The dependence of these five results is summarised by



Here the last three results all use estimates for the same lattice family and the same continuum hierarchy supplied by the first two. The arrows show the order of construction, not a claim that compactness alone gives their analytic inputs.

For each gauge group, we make one joint choice of the family in Proposition 5.1. For each prescribed increasing volume sequence, we then make one joint choice of a refinement and hierarchy for Propositions 5.2–5.5. Their conclusions are not asserted for an arbitrary regular family or an arbitrary smooth table of correlations. With these common choices, the deduction of Theorem A is just the assembly of the five conclusions; it appears at the end of the next section.

5 The principal propositions

The five propositions below use the joint choices described in Section 4. Their complete proofs belong to Part II; the explanations here give the ideas needed to follow the deduction of Theorem A.

Removing the lattice spacing. At a fixed physical volume, increasing the number of sites must not change what the model looks like at a fixed larger distance. The coupling and the normalisation of each observable are chosen to achieve this compatibility. The first proposition gives both one such choice and convergence along its entire spacing sequence.

Proposition 5.1 (Ultraviolet construction)**.** *For every gauge group with simple* $\mathfrak{g}$ *(Definition 2.1), there exists a regular Wilson family* $\mathcal{W}$ *(Definitions 2.7 and 2.8) such that, for every* $p, n \in \mathbb{N}$ *and*

$v \in \mathcal{T}^n$, there exists a smooth kernel $K_{p,v}$ on the periodically separated set of Definition 2.13 with

$$\lim_{N\to\infty} C_{N,p}(v;f) = \int f(x) K_{p,v}(x)\,dx \tag{5.1}$$

for every separated test f supported inside volume p (Definitions 2.9, 2.10 and 2.11).

Idea of the proof.

Step 1. Integrate the fine variables into a density for the coarser variables, with its total integral preserved: Here ρ_j is the effective density at scale j, and U, U' denote the fine and coarse link variables.

$$\int \rho_{j+1}(U')\,dU' = \int \rho_j(U)\,dU.$$

Step 2. Split the estimate according to whether the plaquette matrices are within the small-field threshold. Near the identity the quadratic energy gives a Gaussian comparison; outside that region the energy cost bounds the contribution rather than deleting it.

Step 3. Repeating the integration changes the coefficient of squared curvature. The bare coupling is chosen to compensate for that change, with the logarithmic running required by Reg1.

Step 4. Carry each insertion through the same integrations and normalise its coefficients at the fixed reference level:

$$I_{N,p} = \frac{1}{a_N^d} P_d\left[S_{\theta_N^{-1}} \tau_{N,p,d}^{-1} v\right],$$

as in (6.23). This compares different cutoffs using the same reference coefficients.

Step 5. Stop at an intermediate scale. First make the remaining fine-scale error small by choosing that scale, then take the cutoff limit at the fixed intermediate scale. The two errors give the Cauchy criterion for the full cutoff sequence.

Unresolved. The later ultraviolet sections must prove the estimates for integrating small and large fields, identify the change of the actual effective coupling, and establish the fixed-scale Cauchy estimate with a uniformly small remaining error. The fixed-volume conclusion has been checked in the formal source; its analytic proof is still missing from this manuscript.[1]

Enlarging the volume and separating phases. The estimates for a fixed collection of observables must remain bounded when the surrounding box grows. They give compactness, but the limit can retain several possible vacuum states. The next proposition includes the separate argument that resolves this average into phases.

Proposition 5.2 (Continuum limit and phases)**.** *For the jointly chosen family of Proposition 5.1 and every strictly increasing $p : \mathbb{N} \to \mathbb{N}$, there exist a strictly increasing $q : \mathbb{N} \to \mathbb{N}$ and hierarchy data (Definition 2.17) satisfying the limit P1 and the phase and regularity conditions R7–R12 of Definition 2.18.*

Idea of the proof.

[1] Formal correspondence at revision `3456e5e`: `Construction.lean`, lines 15–40; `HierarchyBridge.lean`, lines 16–26; `YM/FullVolume/Profiles.lean`, lines 25–47..

Step 1. Reflection positivity and the ultraviolet cell bounds of Definitions 8.2 and 8.6 control a joint observation by the single-observation roots:

$$\left|\left\langle \prod_C \Theta_C F_C \right\rangle\right| \le \prod_C \|F_C\|_{\mathcal{C}}.$$

The resulting bounds do not grow with the surrounding box.

Step 2. Slice integration bounds the change in a correlation when one insertion moves. Boundedness and this uniform continuity select a common subsequence for the polynomial basis tuples; summation by parts then identifies derivatives with higher-degree insertions.

Step 3. The full fixed-volume limits from Proposition 5.1 identify the inner limits. Only the outer volume refinement remains, in the order of (2.13).

Step 4. Reflection positivity builds a Hilbert space with nonnegative energy operator H; its evolution projects onto the zero-energy subspace P_0:

$$e^{-tH} \xrightarrow[t\to\infty]{} P_0.$$

Insertions between these projections give operators on the zero-energy subspace. Rotation identities make those operators commute.

Step 5. Represent the commuting operators as functions on the compact phase space. Their vacuum expectations define the probability measure and the mixed correlations:

$$S^{\mathrm{mix}}(a;x) = \int_X S^{\xi}(a;x)\,\nu(d\xi).$$

Unresolved. The infinite-volume argument still needs the same-family bounds for local observables and their derivative representatives from the ultraviolet sections. The later symmetry and phase sections must also prove that the contribution from a far-away stress integral tends to zero and that the long-time operators commute; neither follows from compactness alone.[2]

Recovering relativistic fields. Euclidean time evolution damps positive energies. Allowing time to become complex relates this damping to unitary time evolution, and hence to relativistic vacuum expectations. The estimates must hold as the imaginary time separations tend to zero, because the resulting fields are distributions.

Proposition 5.3 (Relativistic reconstruction)**.** *For the jointly chosen hierarchy of Proposition 5.2, the selected theory satisfies R1–R6 and the continuation satisfies R13–R15 of Definition 2.18, with the same fields and phase as in Definition 2.17.*

Idea of the proof.

Step 1. Smear Euclidean insertions in separated positive-time intervals and let their widths decrease. The uniform bounds give distributional limits for the ordered products.

[2]Formal correspondence at revision `3456e5e`: `YM/Extraction/Limits.lean`, lines 26–43; `HierarchyBridge.lean`, lines 44–57 and 82–119; the phase representation is in `YM/Phase/Mixture.lean`..

Step 2. Positivity makes these limits the Osterwalder–Schrader inner products of positive-time field words F, G, where ΘF is the reflected conjugate:

$$\langle A, B\rangle_{\mathrm{OS}} = S(\Theta A\, B),$$

with words from (2.18). Complete their span and let a further smeared field act by left multiplication.

Step 3. Continue the Euclidean rotation identities to Lorentz transformations. Positivity of Euclidean energy then gives the forward-cone spectrum; equality of the two analytic orderings at spacelike separation gives locality.

Step 4. Quotient by the coefficients whose correlations all vanish, as in Proposition 8.13. The quotient supplies faithful field labels, and the constant observable supplies a nonzero field.

Unresolved. The reconstruction section must prove the bounds as imaginary time separations vanish and justify the equality of the two analytic orderings at spacelike separation. The exact selected-phase and product-test conclusions have been checked in the formal source; these analytic arguments remain unwritten.[3]

Replacing a cluster by a single field. A distant observation cannot resolve every detail of several nearby insertions. The operator product expansion expresses their effect through a finite list of fields at the cluster centre. The coefficients must be independent of which distant observations are subsequently made.

Proposition 5.4 (Common operator product expansion)**.** *The fields and coefficients in H7 of the jointly chosen hierarchy satisfy O1–O2 of Definition 2.20, for the clusters of Definition 2.19.*

Idea of the proof.

Step 1. Bring a fixed cluster of nearby insertions to its centre and expand it in the chosen fields, using one coefficient system for all later distant insertions.

Step 2. Keep degrees at most D. The first omitted degree controls the error, with an arbitrarily small loss to absorb logarithmic factors:

$$\left| S^s(a,x;b,y) - \sum_{j=0}^{D} \sum_{\kappa\in \mathrm{Ch}_j} c_{j,\kappa}(a;x,o) S^s\big((j,O_{j,\kappa}),o;b,y\big) \right| \le r^{D+1-w(a)-\varepsilon},$$

the estimate of (2.31).

Step 3. Choose coefficients from a fixed collection of distant-field correlations before choosing the spectators or phase. The construction proves the error bound in operator norm; evaluating in any phase or in the average state preserves that bound.

Unresolved. The product section must prove the local-product remainder estimate and show how the fixed coefficient choice gives an operator bound before evaluation in a phase. The quantifiers and both powers of the radius in O1–O2 have been checked; their analytic proof is still deferred.[4]

[3] Formal correspondence at revision `3456e5e`: `PublicTheory.lean`, lines 93–319; `HierarchyBridge.lean`, lines 82–133; boundary and locality proofs are organised in `YM/Work/C01.lean` through `C29.lean`..

[4] Formal correspondence at revision `3456e5e`: `ChallengeSpec.lean`, lines 58–80; `HierarchyBridge.lean`, lines 68–80; `YM/OPE/Orbits/Kernel.lean`, lines 62–77..

Detecting the logarithmic correction. The construction supplies a stress tensor and subtracts the selected vacuum expectation from its trace. Comparison at short distances identifies the leading two-point singularity of this centred field. The logarithmic correction to that singularity is incompatible with the finite free-field expressions in Definition 2.22.

Proposition 5.5 (Trace law and finite-profile exclusion)**.** *The field $\Theta \in F_4$ of the jointly chosen hierarchy satisfies T1–T4 of Definition 2.23 in the theory of Proposition 5.3, with the two-point function of Definition 2.21.*

Idea of the proof.

Step 1. Compare the short-distance trace insertions, with their prescribed normalisations, to free Maxwell curvature fields of Section 9. Each trace insertion contributes one inverse running coupling, so the two-point function has two such factors.

Step 2. The logarithmic growth of the coupling therefore gives the scale

$$\lim_{r\downarrow 0} r^8 \left(\log \frac{1}{r}\right)^2 G_\Theta(r) = \frac{3 \dim_{\mathbb{R}} \mathfrak{g}}{\pi^4},$$

as in (2.35); the running coefficient and trace normalisation both enter this amplitude.

Step 3. First the comparison holds on selected radii. The positive-energy representation

$$G_\Theta(r) = \int_{[0,\infty)} e^{-rE}\, d\sigma(E), \qquad d\sigma \geq 0,$$

makes G_Θ decreasing in r. Values at nearby selected radii squeeze the value at each intermediate radius, giving the full right limit.

Step 4. A finite Wick profile has only the powers and logarithms allowed in Definition 2.22; its small-radius expansion cannot agree arbitrarily close to zero with the positive inverse-logarithmic law.

Unresolved. The trace section must prove the stress normalisation, comparison along subsequences of arbitrary radii, the spectral comparison between nearby separations, and the small-radius expansions of the finite comparison kernels (2.33). The amplitude and the exact finite comparison class have been checked; these proofs remain to be supplied.[5]

Deduction of Theorem A from the principal propositions. Fix the gauge group and choose the family of Proposition 5.1, including its insertions and positive regularity slope. For a prescribed strictly increasing p, take the common q and hierarchy of Proposition 5.2. This gives P1 and R7–R12 on that refinement. Proposition 5.3 supplies the remaining reconstruction properties, hence P2. Propositions 5.4 and 5.5 give P3 and P4 for those same objects. □

[5]Formal correspondence at revision `3456e5e`: `TraceBridge.lean`, lines 14–44; `YM/TraceExtension/Verse/Cape.lean`, lines 53–88; `YM/TraceExtension/Wick/Log.lean`, lines 96–108..

Part II
The construction

Part I states what the limiting theory must satisfy. Here we explain how to construct it from the Wilson measure of Definition 2.4. The first task is to compare the same physical observation on successively finer lattices. The second is to show that its correlations remain controlled when the surrounding box grows. Only after these two tasks can we ask how the limit describes fields, symmetries and phases.

The construction has five stages:

1. Choose local link polynomials and normalise each insertion at a fixed reference level, as in (6.23).
2. Integrate successively finer variables while preserving the probability integral; prove the fixed-volume limit and the bounds independent of volume needed for extraction.
3. Select one common refinement giving the ordered limits (2.13) for every list of observable indices and every separated test.
4. Use reflection positivity to build the Hilbert space, establish rotations, and only then represent the long-time operators by phase functions, giving (2.24).
5. On this same hierarchy, construct the selected relativistic theory and prove the common product estimates and the trace law.

Following an observable across scales. Section 6 begins with the exact finite probability model. Its symmetries hold before any limiting argument. It then describes a local observable by a gauge-invariant polynomial in the Taylor coefficients of a classical connection (Definition 6.4), and represents that polynomial using finitely many lattice links. The coefficients of this representative change when short-distance variables are integrated out. Fixing its coefficients at a reference scale lets us follow the same observable index through different fine lattices, as in Definition 2.6 and (6.23).

Section 7 constructs the local polynomial representatives and organises products of insertions so that their coefficients can be followed through integration. The density expansions also record which lattice regions contribute and which plaquettes have large deviations. This organisation prepares the estimates; it does not itself prove them.

The later ultraviolet sections perform the integrations. Averaging over finer variables produces a probability density for coarser variables; the averaging is normalised so that it preserves the original integrals. The difficulty is to control the density and the observables after this operation has been repeated many times. Configurations with plaquette matrices close to the identity admit comparison with an integral having quadratic energy. The complementary configurations must be bounded using their energy cost. Local contributions are organised by the regions of the lattice on which they depend, so that estimates for a fixed local observable can be made independent of the surrounding volume. The change in the coefficient of squared curvature is included in the running coupling.

These arguments have two different uses. At a fixed box size they must prove that the correlations are Cauchy along the entire spacing sequence, as in Proposition 5.1. They must also give bounds on local observables and their difference quotients independent of the box size. The latter bounds permit extraction of an infinite-volume limit; the former identify its inner limits with the full

fixed-volume limits. For the moments of Definition 2.9, the intended passage is the ordered limit (2.13):

$$C_{N,p}(v;f) \longrightarrow R_p \qquad (N \to \infty,\ p \text{ fixed}),$$
$$R_{p(q(k))} \longrightarrow \int f(x) S^{\mathrm{mix}}(\pi(v_1),\dots,\pi(v_n);x)\,dx \quad (k \to \infty).$$

Here f is a separated test of Definition 2.10; the inner limit is required at every volume containing its support (Definition 2.11). The family is chosen before the prescribed volume sequence p; the refinement q is common to all observables and tests.

From correlation functions to fields. Section 8 uses reflection positivity to bound correlations by products of reflected local observables filling the torus. The volume-independent estimates then give a common subsequence for all correlations. To identify derivatives, one compares lattice differences with higher-degree observables. Coefficients invisible in every correlation are discarded, giving the field spaces of the limit. Thus smoothness requires estimates for these comparisons as well as compactness. The spacing refinement introduced during extraction is removed using the fixed-volume theorem, leaving exactly the ordered limit of Definition 2.12.

Section 9 compares these fields at short distances with free Maxwell curvature fields. It supplies finite tests for distinguishing field classes and comparison maps for the later stress-tensor argument. These maps require short-distance estimates for the chosen family and a separate dimension bound. This comparison does not identify the resulting interacting theory with a free theory. The symmetry argument also needs a Hilbert space built from the average correlations: reflection positivity turns products supported at positive times into vectors, with reflected correlations as their inner products. In that space, integral identities for the stress tensor will give continuous rotations. This is why the construction of that Hilbert space precedes the rotation argument, while the decomposition into phases comes afterward.

Phases and short distances. Long-time evolution suppresses positive energies and leaves the zero-energy vectors. Observables placed between long evolutions therefore act on this zero-energy subspace. The rotation identities are used to prove that the resulting operators commute. These operators can then be represented as continuous functions on the compact phase space of Definition 2.17, with expectation given by the measure in (2.24). A selected phase is continued in time to obtain the relativistic fields of Proposition 5.3. The trace and product sections return to the short-distance estimates for the same construction: the former prove the full-radius logarithmic law, and the latter choose one system of product coefficients valid in every phase.

Reading the construction. On a first reading, follow the lattice law and the reference normalisation in Section 6, then the passage from bounded correlations to smooth fields in Section 8. The intervening ultraviolet arguments explain why those bounds hold for one chosen family. The later Maxwell, symmetry, phase and reconstruction arguments explain what additional information is needed to turn that limit into the hierarchy of Theorem A. For each quantitative estimate, keep track of what is fixed, what tends to a limit, and whether its constants depend on the volume. The proofs must use the same family and the same retained correlations throughout.

6 Lattice fields

The field at a point will be represented by observables on finitely many nearby links. There are two choices to make: which local quantity an observable represents, and how to change its coefficients as the lattice spacing decreases. This section specifies these choices without yet integrating out any scales.

We begin with the exact symmetries of the Wilson measure in Subsection 6.1. Subsection 6.2 describes classical local observables by polynomials in Taylor coefficients and chooses finite bases for them. Subsection 6.3 states how a polynomial in link matrices must reproduce these classical observables. Subsection 6.4 then fixes the coefficients of the lattice insertions at a reference scale. Section 7 constructs the local polynomials and introduces the activity sums used in the integration argument; the changes of coefficients between scales require the later renormalisation steps.

We use the lattice, plaquettes and measure of Definitions 2.2, 2.3 and 2.4. The gauge group is as in Definition 2.1; simplicity and connectedness are not needed in this section.

6.1 The Wilson measure

The action depends on the transport around each elementary square, or plaquette, as explained in Definition 2.3. Conjugating that transport or reversing its direction leaves the action unchanged. This observation gives exact symmetries of the probability measure, before any limit is taken.

Definition 6.1 (Walk transport)**.** A walk w records a starting site x_0 and a finite ordered list of steps $(\sigma_j, \mu_j)_{j=1}^{r}$, with $\sigma_j \in \{-1, 1\}$ and $1 \le \mu_j \le 4$. Set $x_j = x_{j-1} + \sigma_j e_{\mu_j}$. For the links of Definition 2.2, its transport is the ordered product

$$U(w) = W_1 \cdots W_r, \qquad W_j = \begin{cases} U(x_{j-1}, \mu_j), & \sigma_j = 1, \\ U(x_j, \mu_j)^{-1}, & \sigma_j = -1. \end{cases} \tag{6.1}$$

The empty walk has transport 1.

For a walk from x to z, the intermediate gauge matrices cancel in this product. The gauge action of Definition 2.5 gives

$$U^{\gamma}(w) = \gamma(x) U(w) \gamma(z)^{-1}. \tag{6.2}$$

For the estimates below, abbreviate the single-plaquette action and define its associated distance from the identity by

$$s(u) = 1 - \frac{1}{m} \operatorname{Re} \operatorname{tr} u, \qquad \chi(u) = \frac{\|u - 1\|_{\mathrm{F}}}{\sqrt{m}} \qquad (u \in G), \tag{6.3}$$

where the Frobenius norm is given by $\|M\|_{\mathrm{F}}^2 = \operatorname{Re} \operatorname{tr}(M^* M)$ and M^* is the conjugate transpose. For $u, v \in G$, unitarity gives

$$\chi(u)^2 = 2 s(u), \qquad 0 \le s(u) \le 2, \qquad \chi(uv) \le \chi(u) + \chi(v). \tag{6.4}$$

Indeed, $\|u - 1\|_{\mathrm{F}}^2 = 2m - 2 \operatorname{Re} \operatorname{tr} u$ and $\|u - 1\|_{\mathrm{F}} \le 2\sqrt{m}$. For the last inequality, write $uv - 1 = (u - 1) + u(v - 1)$ and use invariance of the Frobenius norm under unitary multiplication. Both s and χ are continuous and invariant under conjugation and inversion.

The coordinate symmetries we use are the permutations and reversals of the four lattice axes. They act on link configurations by transporting each link along its image.

Definition 6.2 (Coordinate symmetries)**.** Let H_4 be the group of signed permutation matrices, with one nonzero entry ± 1 in each row and column. For $A(y) = hy + z$, with $h \in H_4$ and $z \in \Lambda_\ell$, the pullback A^*U reads the transport along the image of each positive link. A reversed image uses the inverse variable, with its initial vertex shifted accordingly. The lattice and link conventions are those of Definition 2.2.

Write $\mathcal{U}_\ell$ for the configuration space of Definition 2.2, $C(\mathcal{U}_\ell; \mathbb{C})$ for its continuous complex functions, and $B_b(\mathcal{U}_\ell; \mathbb{C})$ for its bounded Borel complex functions. Here $L^1(\mu_{\ell,\beta})$ denotes the integrable functions for the Wilson measure of Definition 2.4. The following proposition justifies ordinary expectations of local polynomials and the changes of variables used throughout the construction.

Proposition 6.3 (The lattice law)**.** *For $\ell \geq 1$ and $\beta \in \mathbb{R}$, the measure of Definition 2.4 satisfies*

$$\mu_{\ell,\beta}(\mathcal{U}_\ell) = 1, \qquad C(\mathcal{U}_\ell; \mathbb{C}) \cup B_b(\mathcal{U}_\ell; \mathbb{C}) \subseteq L^1(\mu_{\ell,\beta})\,. \tag{6.5}$$

For every gauge transformation γ of Definition 2.5, coordinate symmetry A of Definition 6.2, and $F \in L^1(\mu_{\ell,\beta})$,

$$\int F(U^\gamma)\, d\mu_{\ell,\beta}(U) = \int F(U)\, d\mu_{\ell,\beta}(U) = \int F(A^*U)\, d\mu_{\ell,\beta}(U)\,. \tag{6.6}$$

Proof. There are three checks: the density is finite and positive, each change of variables preserves product Haar measure, and each preserves the action. Together they preserve the probability law. Write η_ℓ for the product Haar measure in (2.4).

1. *Normalisation and integrability.* The closed group $G \subset U(m)$ and its finite configuration space are compact. There are six plaquettes per site, each with energy in $[0, 2]$. Thus

$$0 \leq E_\ell \leq 12\ell^4, \qquad e^{-12|\beta|\ell^4} \leq e^{-\beta E_\ell(U)} \leq e^{12|\beta|\ell^4}\,. \tag{6.7}$$

   Integrating against probability Haar gives a finite positive normalising constant $Z_{\ell,\beta}$ as in (2.4). A continuous function on the configuration space is bounded, so both classes in (6.5) are integrable.

2. *Reference measure.* Haar invariance allows a fixed matrix to be multiplied on either side of a link, and allows its orientation to be reversed. Let h be the probability Haar measure of Definition 2.4; T_*h denotes its image under a map T. For $R_g(u) = ug$, $\mathrm{inv}(u) = u^{-1}$ and every integrable $f : G \to \mathbb{C}$, the identities are

$$(R_g)_*h = h, \qquad \mathrm{inv}_*h = h, \qquad \int_G f(aub)\, h(du) = \int_G f(u)\, h(du) \quad (a, b \in G)\,. \tag{6.8}$$

   Indeed, $(R_g)_*h$ is left invariant with mass one, so Haar uniqueness identifies it with h. The image under inversion is then left invariant by right invariance of h, again with mass one. Applying these identities independently to each link gives, for the gauge map $\Gamma_\gamma : U \mapsto U^\gamma$,

$$(\Gamma_\gamma)_*\eta_\ell = \eta_\ell, \qquad (A^*)_*\eta_\ell = \eta_\ell\,. \tag{6.9}$$

   For the second identity, the coordinate symmetry permutes the positive links and inverts those whose image orientation is negative.

3. *Action.* By (6.2), a gauge transformation conjugates a plaquette transport at its initial vertex. A coordinate symmetry sends an elementary square to another one, up to a change of initial corner and orientation. To check the counting, write $he_\mu = \sigma_\mu e_{\pi(\mu)}$, with $\sigma_\mu \in \{-1, 1\}$. Write $\mathbf{1}_{\{P\}}$ for 1 when the condition P holds and 0 otherwise. The positive-axis base of the image square is
$$hy + z - \mathbf{1}_{\{\sigma_\mu=-1\}} e_{\pi(\mu)} - \mathbf{1}_{\{\sigma_\nu=-1\}} e_{\pi(\nu)},$$
and its axis pair is the increasing rearrangement of $\{\pi(\mu), \pi(\nu)\}$. This permutes the face indices even when $\ell = 1$ or 2. Since conjugation and inversion preserve s, we have
$$E_\ell(U^\gamma) = E_\ell(U), \qquad E_\ell(A^*U) = E_\ell(U)\,. \tag{6.10}$$
Thus the transformed density agrees pointwise with the original density:
$$e^{-\beta E_\ell(TU)} = e^{-\beta E_\ell(U)}, \qquad T \in \{\Gamma_\gamma, A^*\}.$$
For either $T = \Gamma_\gamma$ or $T = A^*$, equations (6.9) and (6.10) now give
$$\int F(TU)\, d\mu_{\ell,\beta}(U) = \frac{1}{Z_{\ell,\beta}} \int F(TU) e^{-\beta E_\ell(TU)}\, d\eta_\ell(U) = \int F(U)\, d\mu_{\ell,\beta}(U)\,.$$
This proves (6.6) for every integrable F.

□

6.2 Classical polynomials

We now specify the classical quantities that the lattice insertions of Definition 2.6 will represent. A classical connection is a smooth $\mathfrak{g}$-valued one-form $A = \sum_\mu A_\mu(z)\, dz^\mu$ near the origin. We describe it in radial gauge: its component in the radial direction vanishes. Its Taylor coefficients then satisfy a linear constraint, degree by degree. The constant coefficient vanishes: a constant one-form with zero contraction against every radial vector must be zero. A polynomial in finitely many such coefficients describes a classical local observable; invariance under simultaneous conjugation removes the choice of gauge at the origin.

Definition 6.4 (Gauge-invariant polynomials)**.** For $\mathfrak{g}$ of Definition 2.1 and $j \in \mathbb{N}$, let $\mathcal{A}_j$ be the real space of homogeneous polynomial one-forms of degree $j + 1$, with values in $\mathfrak{g}$, satisfying
$$\sum_{\mu=1}^{4} z^\mu A_{j,\mu}(z) = 0 \qquad (z \in \mathbb{R}^4)\,. \tag{6.11}$$

The list of Taylor coefficients, called a real jet, is a sequence $J = (A_j)_{j\geq 0}$ with $A_j \in \mathcal{A}_j$. Choose real coordinates on each $\mathcal{A}_j$, assigning weight $j + 2$ to each coordinate. The weight of a monomial is the sum of the weights of its factors, counted with multiplicity. For $d \in \mathbb{N}$, let V_d be the space of complex polynomials in these coordinates, of total weight at most d, invariant under simultaneous conjugation $\mathrm{Ad}(g)X = gXg^{-1}$ by $g \in G$.

The coefficients in Definition 6.4 are Taylor coefficients, with the factorial normalisation, rather than derivative values. Equivalently, they are diagonal evaluations of symmetric $(j+1)$-linear maps. The weight $j + 2$ records the scaling of a link integral: the polynomial one-form contributes $j + 1$ powers of length and integration contributes one more. Weights add under multiplication. We also count the number of factors in a monomial, its ordinary polynomial degree; the two gradings will serve different purposes.

Definition 6.5 (Actions and gradings)**.** Let $O(4) = \{h \in M_4(\mathbb{R}) : h^{\mathsf{T}}h = 1\}$ be the group of real orthogonal matrices, where h^{T} is the transpose. An element $h \in O(4)$ acts on the one-forms and invariant polynomials of Definition 6.4 by

$$(T_h A_j)_\mu(z) = \sum_\nu h_{\mu\nu} A_{j,\nu}(h^{-1}z), \qquad (h \cdot v)(J) = v(T_{h^{-1}} J)\,. \tag{6.12}$$

Coefficient conjugation is denoted by $v \mapsto \bar{v}$; a real polynomial has $\bar{v} = v$. Let $V_d^{w,f}$ be the subspace homogeneous of weight w and factor number f. Every monomial of such a polynomial has weight w and ordinary polynomial degree f.

The action (6.12) preserves (6.11) and commutes with conjugation by G.

The weight bound leaves only finitely many coordinates and monomials. We can therefore choose one finite basis at each degree, retaining the basis elements already chosen at lower degrees. Real homogeneous bases make conjugation and the two gradings explicit.

Lemma 6.6 (Finite polynomial spaces)**.** *For the spaces of Definition 6.4, evaluation on real jets is injective, and*

$$\dim_{\mathbb{C}} V_d < \infty, \qquad 1 \in V_d \subseteq V_{d+1}, \qquad V_d = \bigoplus_{\substack{0 \le w \le d \\ 0 \le f \le w}} V_d^{w,f}\,. \tag{6.13}$$

Each summand is preserved by (6.12) *and coefficient conjugation. There are nested real bases, homogeneous in both gradings and containing* 1*.*

Proof. The basis claim follows from three finite-dimensional facts: only finitely many monomials occur, the invariant condition splits by the two grades, and each real homogeneous piece has a real basis.

1. *Finitely many monomials.*

   Only coordinates with $j + 2 \le d$ can occur in a monomial of weight at most d. There are finitely many such coordinates, and the exponent of each is bounded by d. Thus finitely many monomials span the polynomials under consideration, and their invariant subspace is finite dimensional.

2. *Evaluation and homogeneous pieces.* A complex polynomial vanishing on every real jet vanishes identically, by the one-variable polynomial identity theorem applied successively in these finitely many real coordinates.

   Conjugation acts linearly within each $\mathcal{A}_j$ and therefore preserves both grades. Writing $v = \sum_{w,f} v^{w,f}$, comparison of the homogeneous terms gives

   $$v(\mathrm{Ad}(g)J) = v(J) \Longrightarrow \quad v^{w,f}(\mathrm{Ad}(g)J) = v^{w,f}(J) \qquad (g \in G)\,. \tag{6.14}$$

   Thus the homogeneous parts of an invariant polynomial are themselves invariant. Every coordinate has weight at least two; in particular the factor number cannot exceed the weight, giving the index range in (6.13). The same linearity argument applies to (6.12), and these real linear changes commute with coefficient conjugation. Each invariant homogeneous component is the complex span of its real polynomials, by decomposition into real and imaginary coefficient parts.

3. *Nested real bases.* Choose a real basis in each component, beginning with $\{1\}$ for the constant component, and retain these choices as d increases.

□

Definition 6.7 (Chosen polynomial bases)**.** Fix the nested bases of Lemma 6.6. For $d \in \mathbb{N}$, write

$$(b_{d,i})_{0\le i<\dim_{\mathbb{C}} V_d}, \qquad b_{d,0}=1, \qquad b_{d,i}\in V_d^{w_{d,i},f_{d,i}},$$

with the two gradings of Definition 6.5.

We identify each polynomial with its coefficient vector when applying a linear map to V_d. The continuum spaces F_d of Definition 2.17 will arise only after a limiting correlation table and its null relations have been obtained. The spaces V_d contain the classical observable polynomials before these relations are imposed.

6.3 Local representatives

A polynomial in Taylor coefficients is not yet a lattice observable. We seek a polynomial in nearby link matrices whose Taylor expansion, when tested on a classical connection, recovers the prescribed classical observable polynomial. The linear map making this choice for every $v \in V_d$ is called a local section. Its existence is a separate construction; here we state its normalisation precisely.

On $\mathbb{Z}^4$, the gauge action of Definition 2.5 and the coordinate pullbacks of Definition 6.2 use the same formulas as on a torus. Polynomial functions on a finite set of links can then be evaluated on any periodic lattice.

Definition 6.8 (Local polynomial functions)**.** For a finite set $\Sigma \subset \mathbb{Z}^4$, let $\mathcal{P}(\Sigma)$ be the polynomial functions of matrix entries and conjugate entries of links with both endpoints in Σ. For $P \in \mathcal{P}(\Sigma)$ and $o \in \Lambda_\ell$, write $P(o;U)$ for its value after substituting

$$U(x,\mu) \longmapsto U(o+[x]_\ell,\mu)\,. \tag{6.15}$$

Here $[x]_\ell$ is reduction of $x \in \mathbb{Z}^4$ modulo ℓ; the link conventions are those of Definition 2.2.

This implements the finite-site locality requirement Reg4 of Definition 2.8. The substitution is meaningful even when distinct links of $\mathbb{Z}^4$ become identical after reduction modulo ℓ.

To test the classical normalisation, retain the Taylor coefficients that can occur in V_d and form their polynomial connection. Its transport along one edge will be the test link.

Definition 6.9 (Truncated connections and test links)**.** For $d \in \mathbb{N}$, a real jet $J = (A_j)$ of Definition 6.4 and $\varepsilon \in \mathbb{R}$, set

$$\mathcal{J}_d = \bigoplus_{j+2\le d} \mathcal{A}_j, \qquad A^J_\mu(z) = \sum_{j+2\le d} A_{j,\mu}(z), \qquad (\mathrm{sc}_\varepsilon J)_j = \varepsilon^{j+2} A_j\,. \tag{6.16}$$

The finite tuple $(A_j)_{j+2\le d} \in \mathcal{J}_d$ determines A^J; its dependence on d is suppressed in this notation. A finite tuple is viewed as a jet by setting its higher coefficients to zero. For $x \in \mathbb{Z}^4$ and $1 \le \mu \le 4$, define the test link $U_{J,\varepsilon}(x,\mu)$ at spacing ε by the endpoint of

$$\begin{aligned} B(t) &= \varepsilon A^J_\mu\big(\varepsilon(x+te_\mu)\big), & u'(t) &= u(t)B(t),\\ u(0) &= 1, & U_{J,\varepsilon}(x,\mu) &= u(1)\,. \end{aligned} \tag{6.17}$$

Here $t \in [0,1]$ parametrises the edge $x + te_\mu$ and $B(t)$ is the matrix coefficient of its transport equation.

We justify existence, uniqueness and analytic dependence of these links before using them. Successive integration gives the ordered series

$$u(t) = 1 + \sum_{r\geq 1} \int_{0<s_1<\cdots<s_r<t} B(s_1)\cdots B(s_r)\, ds_1\cdots ds_r,$$
$$\left\| \int_{0<s_1<\cdots<s_r<t} B(s_1)\cdots B(s_r)\, ds_1\cdots ds_r \right\|_{\mathrm{F}} \leq \frac{M^r}{r!} \tag{6.18}$$

for $r \geq 1$ and $0 \leq t \leq 1$ when $\|B(s)\|_{\mathrm{F}} \leq M$. For a fixed real jet, B is polynomial in ε and the same factorial bound holds uniformly on a small complex ε-neighbourhood of zero. The series is therefore locally uniformly convergent there, solves the equation, and gives an endpoint real analytic in ε near zero. Iterating the integral equation for the difference of two solutions gives the same factorial bound and proves uniqueness. For real ε, step-function approximations to B have solutions that are products of matrices $\exp(\Delta t\, B(t_j)) \in G$. The same series bound proves convergence of these products to u; since G is closed, the links in (6.17) are group valued.

The section must preserve the exact lattice symmetries and the constant observable. Its classical normalisation requires equality of Taylor coefficients through the degree being represented. These are the conditions in the next definition.

Definition 6.10 (Normalised local section)**.** A normalised local section in degree $d \in \mathbb{N}$ is a finite set $\Sigma_d \subset \mathbb{Z}^4$ and a complex linear map $P_d : V_d \to \mathcal{P}(\Sigma_d)$, with V_d from Definition 6.4 and polynomial functions as in Definition 6.8, such that

$$0 \in \Sigma_d, \qquad h\Sigma_d = \Sigma_d \quad (h \in H_4), \tag{6.19}$$

and, for all $v \in V_d$, link configurations U on $\mathbb{Z}^4$, $\gamma : \mathbb{Z}^4 \to G$ and $h \in H_4$,

$$\begin{aligned} P_d[v](U^\gamma) &= P_d[v](U), & P_d[1] &= 1, \\ P_d[\bar{v}](U) &= \overline{P_d[v](U)}, & P_d[v](h^*U) &= P_d[h\cdot v](U)\,. \end{aligned} \tag{6.20}$$

For every real jet J and integer $0 \leq k \leq d$, it also satisfies

$$\frac{d^k}{d\varepsilon^k} P_d[v](U_{J,\varepsilon})\Big|_{\varepsilon=0} = \frac{d^k}{d\varepsilon^k} v(\mathrm{sc}_\varepsilon\, J)\Big|_{\varepsilon=0}\,. \tag{6.21}$$

The jets and their coordinate action are as in Definitions 6.4 and 6.5; H_4 and its pullbacks are from Definition 6.2, and the scaling and test links are from Definition 6.9.

The Taylor identity imposes no equality at nonzero spacing. No compatibility of P_d and P_{d+1} is required. Proposition 7.2 constructs these sections and proves the derivative estimates needed for the later blocking argument.

6.4 Renormalised insertions

As shorter scales are integrated out, the coefficients of an observable change and different classical observable polynomials can mix. We normalise the observable by its coefficients at one reference level, chosen independently of cutoff and volume. The coefficient transfer is the linear map from fine-lattice coefficients to those at the reference level. Its inverse determines which fine-lattice polynomial represents a prescribed polynomial at that reference level. This describes a particular choice of the insertions in Definition 2.6, not an additional condition on that definition.

Use a_N, N, p from Definition 2.7, the basis from Definition 6.7, and a local section from Definition 6.10. In the terminology of Definition 2.6, the observable index will be (d, v), where $v \in V_d$ is the classical polynomial and d its assigned scaling degree.

Definition 6.11 (Reference normalisation)**.** For each degree d, choose a reference level $r(d) \in \mathbb{N}$, independently of N, p. Let $\theta_N > 0$ be the threshold that will distinguish small plaquette distances χ of (6.3) in the later estimates. Define the linear factor scaling on the basis of Definition 6.7 by

$$S_t b_{d,i} = t^{f_{d,i}} b_{d,i} \qquad (t > 0)\,. \tag{6.22}$$

For an invertible complex linear map $\tau_{N,p,d} : V_d \to V_d$, the candidate insertion is

$$I_{N,p}\big((d, v), o; U\big) = \frac{1}{a_N^d} P_d \Big[S_{\theta_N^{-1}} \tau_{N,p,d}^{-1} v \Big]\,(o; U), \tag{6.23}$$

with placement (6.15).

Factor scaling multiplies a component with f factors by $\frac{1}{\theta_N^f}$; the prefactor $\frac{1}{a_N^d}$ supplies the physical degree normalisation.

For $N \geq r(d)$, let $x_j : V_d \to V_d$ be complex linear maps from coefficients at the finest level N to coefficients at level j, for $r(d) \leq j \leq N$. The conditions that make $\tau_{N,p,d}$ a reference transfer are

$$x_N(v) = v, \qquad x_{r(d)}(v) = \tau_{N,p,d} v, \qquad x_j(a_N^d 1) = a_j^d 1 \quad (r(d) \leq j \leq N)\,. \tag{6.24}$$

These equalities fix how coefficients are normalised between levels; they assert no pointwise identity between observables on different lattices. Reality and signed-permutation covariance require

$$\tau_{N,p,d} \bar{v} = \overline{\tau_{N,p,d} v}, \qquad \tau_{N,p,d}(h \cdot v) = h \cdot (\tau_{N,p,d} v) \quad (h \in H_4)\,. \tag{6.25}$$

The locality of the resulting insertion follows from its polynomial formula, even on a torus smaller than the chosen set of sites. The unit normalisation is different: it uses the reference-transfer conditions and therefore applies only beyond the reference level. The following proposition separates these two conclusions.

Proposition 6.12 (Locality of the insertions)**.** *Let P_d satisfy Definition 6.10, $\theta_N > 0$ and $\tau_{N,p,d} \in \mathrm{GL}_{\mathbb{C}}(V_d)$. For all N, p, the map $v \mapsto I_{N,p}\big((d, v), o; \cdot\big)$ in (6.23) is complex linear and satisfies Reg2–Reg4 of Definition 2.8, with finite set Σ_d. Moreover,*

$$I_{N,p}\big((d, v), o; U(\cdot + z)\big) = I_{N,p}\big((d, v), o + z; U\big)\,. \tag{6.26}$$

If $N \geq r(d)$ and (6.24)–(6.25) hold, then

$$\begin{aligned} I_{N,p}\big((d, \bar{v}), o; U\big) &= \overline{I_{N,p}\big((d, v), o; U\big)}, \\ I_{N,p}\big((d, h \cdot v), A(o); U\big) &= I_{N,p}\big((d, v), o; A^* U\big), \\ I_{N,p}\big((d, a_{r(d)}^d 1), o; U\big) &= 1, \qquad x_{r(d)}(\tau_{N,p,d}^{-1} v) = v \end{aligned} \tag{6.27}$$

for $A(o) = ho + z$, $h \in H_4$, as in Definition 6.2.

Proof. The conclusions follow in three steps: the defining polynomial gives regularity and locality, the symmetries of its factors give covariance, and the transfer endpoint fixes the unit.

1. *Locality and continuity.*

   The coefficient changes in (6.23) are complex linear. A polynomial in link entries and their conjugates is continuous, and the section is gauge invariant by (6.20); this proves Reg2 and Reg3. Placement (6.15) reads only links with both endpoints in $o+[\Sigma_d]_{L_pD_N}$, proving Reg4. Identifying vertices of $\mathbb{Z}^4$ after reduction modulo L_pD_N does not enlarge that set. Translating the placement proves (6.26). In particular, the dependencies are

$$I_{N,p}((d,v),o;U) = \frac{1}{a_N^d} P_d[S_{\theta_N^{-1}} \tau_{N,p,d}^{-1} v](o;U).$$

2. *Reality and coordinate covariance.* Factor scaling commutes with coefficient conjugation and H_4, since both actions preserve factor number by Lemma 6.6. Together with (6.25) and (6.20), this gives the first two identities in (6.27).

3. *Reference unit.* For the unit, the endpoint conditions (6.24) give

$$\begin{aligned} \tau_{N,p,d}^{-1}(a_{r(d)}^d 1) &= a_N^d 1, \\ I_{N,p}\big((d, a_{r(d)}^d 1), o; U\big) &= \frac{1}{a_N^d} P_d\big[S_{\theta_N^{-1}}(a_N^d 1)\big](o;U) = \frac{a_N^d}{a_N^d} P_d[1](o;U) = 1\,. \end{aligned} \tag{6.28}$$

   The remaining identity follows by applying the reference endpoint condition to $\tau_{N,p,d}^{-1} v$.

□

The set of observable indices in Definition 2.6 is therefore $\mathcal{T} = \bigsqcup_d V_d$ for this construction. The disjoint union retains the degree d, even if v belongs to a smaller polynomial space. Under the hypotheses of Proposition 6.12, the requirements Reg2–Reg4 are satisfied at every cutoff; the reference unit is $a_{r(d)}^d 1$. Proposition 6.3 makes every lattice moment in (2.10) an integral of a continuous, hence integrable, product of observables. The running-coupling condition Reg1 and convergence of these moments require the later ultraviolet arguments.

Unresolved. The invertible coefficient transfers satisfying (6.24)–(6.25) must still be constructed for the same Wilson family. Proposition 6.12 assumes them. The local sections and their derivative bounds are supplied by Proposition 7.2; transferring their coefficients through the integration over scales remains a later ultraviolet proof.[6]

7 Local polynomials and activities

Section 6 specified which classical observable a lattice polynomial must reproduce at small spacing. We now construct those polynomials and introduce the form in which local interactions will be carried through the integration over scales. The common principle is to retain the region and the coefficients of every term: this allows multiplication and summation without losing the estimates on individual observables.

Subsection 7.1 recovers finitely many Taylor coefficients from finitely many transported plaquettes. Subsection 7.2 turns this recovery into the sections of Definition 6.10, with derivative bounds independent of the torus size. Subsection 7.3 records insertions as coefficients of auxiliary commuting

[6] Formal correspondence at revision `3456e5e`: Transfer conditions: `YM/Source/Insertion.lean`, lines 89–114; same-family application: `YM/FullVolume/Cross.lean`, lines 30–71.

variables whose squares vanish. Subsection 7.4 defines the local activity bounds and proves absolute convergence of the resulting sums. Constructing the coefficient transfers and proving the bounds after an integration step remain later tasks.

7.1 Recovering Taylor coefficients

A plaquette transport has, to first order near the zero connection, the curvature averaged over its square. Different squares therefore give different linear measurements of the Taylor coefficients of a connection. For any fixed degree, finitely many such measurements suffice; radial gauge is what makes recovery of the connection from its curvature possible.

Fix $d \in \mathbb{N}$. Use the spaces $\mathcal{A}_j, V_d$ of Definition 6.4 and the finite tuples $J \in \mathcal{J}_d$ of Definition 6.9. Such a tuple specifies the polynomial connection A^J and the test links $U_{J,\varepsilon}$ defined there. On $\mathfrak{g}$ use the norm $|X|_{\mathfrak{g}} = \frac{\|X\|_{\mathrm{F}}}{\sqrt{m}}$, and let $P_{\mathfrak{g}}$ be real Frobenius orthogonal projection onto $\mathfrak{g}$. The following matrix-valued function extracts the leading curvature; here $u \in G$, and χ is the distance from the identity in (6.3):

$$\mathcal{F}(u) = P_{\mathfrak{g}}\left(\frac{u - u^{-1}}{2}\right), \qquad |\mathcal{F}(u)|_{\mathfrak{g}} \le \chi(u), \qquad \frac{d}{ds}\mathcal{F}(e^{sX})\Big|_{s=0} = X \quad (X \in \mathfrak{g}). \tag{7.1}$$

Projection commutes with conjugation by G, since conjugation is orthogonal and preserves $\mathfrak{g}$. The bound follows by writing $u - u^{-1} = (u-1) - (u^{-1}-1)$ and using (6.4). The derivative follows from $e^{sX} - e^{-sX} = 2sX + O(s^3)$. Also $\mathcal{F}(u^{-1}) = -\mathcal{F}(u)$.

For a plaquette p based at $x \in \mathbb{Z}^4$ and a walk ω from 0 to x, transport its curvature back to the origin:

$$Q_{\omega,p}(U) = \mathrm{Ad}\big(U(\omega)\big)\mathcal{F}(U_p), \qquad Q_{\omega,p}(U^\gamma) = \mathrm{Ad}\big(\gamma(0)\big)Q_{\omega,p}(U). \tag{7.2}$$

Here $U(\omega)$ is the ordered walk transport of Definition 6.1, U_p is the plaquette matrix of Definition 2.3, and $\mathrm{Ad}(g)X = gXg^{-1}$ as in Definition 6.4. The second identity uses the gauge transformation of Definition 2.5 and (6.2). All these quantities now transform at the same site, so invariant polynomials in them will be gauge invariant. They read only links in the walk and plaquette, with both endpoints included.

The next lemma supplies a local analytic recovery map. It is a recovery from the joint collection of plaquettes, rather than a logarithm of a single plaquette.

Lemma 7.1 (Finite sampling)**.** *There is a finite family $(\omega_a, p_a)_{a=1}^{M_d}$ for which*

$$\mathcal{Q}_d(J) = \big(Q_{\omega_a,p_a}(U_{J,1})\big)_{a=1}^{M_d}, \qquad \mathcal{Q}_d : \mathcal{J}_d \longrightarrow \mathfrak{g}^{M_d}, \tag{7.3}$$

has injective derivative at 0. There is a real analytic map $R_d : \mathfrak{g}^{M_d} \to \mathcal{J}_d$, defined near 0, satisfying

$$R_d(0) = 0, \qquad R_d\mathcal{Q}_d = \mathrm{id}, \qquad R_d(\mathrm{Ad}(g)q) = \mathrm{Ad}(g)R_d(q) \quad (g \in G). \tag{7.4}$$

The last two equalities hold near the origin, for each fixed $g \in G$; the test links are from Definition 6.9, and $\mathrm{Ad}(g)$ acts on every component of a tuple.

Proof. The recovery has three steps: all square measurements together detect every jet, finite dimensionality reduces them to finitely many measurements, and the inverse function theorem recovers the jet from their nonlinear values. The two properties to establish are

$$\mathcal{Q}_d(J) = \big(Q_{\omega_a,p_a}(U_{J,1})\big)_{a=1}^{M_d}, \qquad R_d\mathcal{Q}_d(J) = J \ \text{ near } 0.$$

1. *All square measurements detect the jet.*

   Recall that $A^J_\mu = \sum_{j+2\le d} A_{j,\mu}$ is the truncated connection of Definition 6.9. Vary its size by $\lambda \in \mathbb{R}$ in the transport equation (6.17), keeping the edge fixed. At $\lambda = 0$ the solution is 1, so its first variation is the integral along that edge:

   $$\frac{d}{d\lambda} U_{\lambda J,1}(x,\mu)\Big|_{\lambda=0} = \int_0^1 A^J_\mu(x + te_\mu)\, dt\,.$$

   Since $\mathcal{F}(1) = 0$, differentiating the conjugating transport in (7.2) contributes zero. Consequently, for the unit square $p = (x;\mu,\nu)$,

   $$D\big(Q_{\omega,p}(U_{J,1})\big)_{J=0}[J] = \int_0^1\int_0^1 (\partial_\mu A^J_\nu - \partial_\nu A^J_\mu)(x + se_\mu + te_\nu)\, ds\, dt\,. \tag{7.5}$$

   This is independent of ω.

   Suppose all these integrals vanish for integer x. Each is a polynomial in x; a polynomial vanishing on $\mathbb{Z}^4$ vanishes everywhere, by successive one-variable polynomial identities. For the highest homogeneous term A_j, the highest part of that integral is $\partial_\mu A_{j,\nu} - \partial_\nu A_{j,\mu}$. It must therefore vanish. Differentiating the radial identity (6.11) and using homogeneity gives

   $$(j+2)A_{j,\mu}(z) = \sum_\nu z^\nu(\partial_\nu A_{j,\mu} - \partial_\mu A_{j,\nu})(z)\,. \tag{7.6}$$

   Thus $A_j = 0$, and descending in degree proves $J = 0$.

2. *Retain finitely many measurements.* The kernels of the linear measurements have zero intersection. While their intersection is nonzero, another measurement decreases its dimension, so finitely many suffice. For $d < 2$ the source space is zero and the assertion is immediate.

3. *Invert the finite nonlinear map.* The map $\mathcal{Q}_d$ is analytic near zero by the transport series (6.18). Set $B = D\mathcal{Q}_d(0)$. Conjugating the connection by a constant $g \in G$ conjugates its test links, by uniqueness of (6.17). Equation (7.2) and differentiation therefore give

   $$\mathcal{Q}_d(\mathrm{Ad}(g)J) = \mathrm{Ad}(g)\mathcal{Q}_d(J), \qquad B\,\mathrm{Ad}(g) = \mathrm{Ad}(g)B\,.$$

   The adjoint action on $\mathfrak{g}^{M_d}$ is orthogonal and preserves $\operatorname{ran} B$ in both directions. Its orthogonal projection Π therefore commutes with that action. Here B^{-1} denotes the inverse from $\operatorname{ran} B$ to $\mathcal{J}_d$. The linear map $C = B^{-1}\Pi$ satisfies

   $$CB = \mathrm{id}, \qquad C\,\mathrm{Ad}(g) = \mathrm{Ad}(g)C, \qquad D(C\mathcal{Q}_d)(0) = \mathrm{id}\,.$$

   Since $\mathcal{Q}_d(0) = 0$, the analytic inverse function theorem gives a local inverse g_d of $C\mathcal{Q}_d$ fixing 0. Then $R_d = g_d C$ has the first two properties in (7.4). The last identity in (7.4) follows from uniqueness of the local inverse, for each fixed $g \in G$.

□

7.2 Constructing the local polynomials

The recovery map is analytic only near the zero connection. Its Taylor polynomial, however, can be evaluated on every link configuration. Composing this polynomial with the classical observable

polynomials $v \in V_d$ of Definition 6.4 gives the required local representatives. A finite average restores the signed coordinate symmetries exactly.

We measure derivatives by varying each link matrix within G. The permitted size of a variation grows with its distance from the insertion site, so we first specify that distance. On the lattice Λ_ℓ of Definition 2.2, periodic maximum distance is

$$d_\ell(x,y) = \min_{z \in \ell\mathbb{Z}^4} \|\widetilde{x} - \widetilde{y} + z\|_\infty \,, \tag{7.7}$$

where tildes denote arbitrary integer representatives and $\|z\|_\infty = \max_\mu |z^\mu|$. For $0 < t \le 1$ and a set of sites $X \subseteq \Lambda_\ell$, define the bad sites and the configurations good on X by

$$\begin{aligned} \mathcal{B}_t(U) &= \{x : \text{some plaquette } p \text{ containing } x \text{ has } \chi(U_p) > t\}, \\ \mathcal{G}_t(X) &= \{U : X \cap \mathcal{B}_t(U) = \varnothing\} \,. \end{aligned} \tag{7.8}$$

Every plaquette touching X is tested, and equality $\chi(U_p) = t$ belongs to the good set. The bad-site indicators are Borel and gauge invariant, by continuity and conjugation invariance of χ.

Fix an insertion site $o \in \Lambda_\ell$ and integers $k, q \in \mathbb{N}$. Here k is the largest derivative order and q controls the permitted growth with distance. A direction A assigns an element of $\mathfrak{g}$ to each positive link e. The path $U(e) \mapsto e^{sA(e)}U(e)$ changes each matrix by multiplication on the left and remains in G for real s (Definition 2.1). Differentiate along this path and impose the bound

$$D_A F(U) = \frac{d}{ds} F\big((e^{sA(e)}U(e))_e\big)\Big|_{s=0}, \qquad |A(e)|_{\mathfrak{g}} \le t\big(1 + d_\ell(e^-, o)\big)^q \,. \tag{7.9}$$

Here e^- is the initial site of the link. Write $\mathrm{Dir}_{t,q}(o)$ for the directions satisfying the bound in (7.9), and define

$$\|F\|_{t,k,q;o,X} = \sup_{\substack{U \in \mathcal{G}_t(X),\ 0 \le r \le k \\ A_1,\dots,A_r \in \mathrm{Dir}_{t,q}(o)}} |D_{A_1} \cdots D_{A_r} F(U)| \,. \tag{7.10}$$

For $r = 0$, the expression under the supremum is $|F(U)|$. This is a seminorm on functions with these derivatives; no value outside $\mathcal{G}_t(X)$ is measured.

The small factor in the following bound counts the number of classical factors, independently of their weights. The section is chosen before the derivative order, the threshold and the torus.

Proposition 7.2 (Normalised sections)**.** *For the bases $b_{d,i}$ and factor numbers $f_{d,i}$ of Definition 6.7, one can choose sections P_d satisfying Definition 6.10, simultaneously for all d. Each $P_d[b_{d,i}]$ is a polynomial in real coordinates of finitely many quantities (7.2), with every monomial containing at least $f_{d,i}$ factors. For every $k, q \in \mathbb{N}$ with $k \ge 6$, $q \ge 2$, there is $A_{d,k,q} \ge 0$ such that*

$$\big\|P_d[b_{d,i}](o;\cdot)\big\|_{t,k,q;o,\,o+[\Sigma_d]_\ell} \le A_{d,k,q}\, t^{f_{d,i}} \qquad (\ell \ge 1,\ o \in \Lambda_\ell,\ 0 < t \le 1) \,. \tag{7.11}$$

Proof. The construction has three parts: approximate the inverse to the required Taylor order, average over signed coordinate changes, and estimate each product of transported plaquettes on the good set. The two key controls are

$$\frac{d^r}{d\varepsilon^r} R_d^{[d]}(\mathcal{Q}_d(\mathrm{sc}_\varepsilon J))\Big|_{\varepsilon=0} = \frac{d^r}{d\varepsilon^r} \mathrm{sc}_\varepsilon J\Big|_{\varepsilon=0} \quad (0 \le r \le d), \qquad \|P_d[b_{d,i}]\|_{t,k,q;o,\,o+[\Sigma_d]_\ell} \le A_{d,k,q} t^{f_{d,i}}.$$

The recovery identity (7.4) makes the Taylor polynomial reproduce the identity through degree d.

1. *Match the Taylor coefficients and retain factors.*

   Let $R_d^{[d]}$ be the ordinary Taylor polynomial at 0 of the recovery map R_d in Lemma 7.1, through degree d. Apply it to the finite tuple of transported plaquettes chosen in that lemma and set

   $$P_d^0[v](U) = v\Big(R_d^{[d]}\big((Q_{\omega_a,p_a}(U))_a\big)\Big)\,. \tag{7.12}$$

   The polynomial $R_d^{[d]}$ has zero constant term, real coefficients, and commutes with conjugation by G, by differentiating (7.4). Hence P_d^0 is gauge invariant, respects coefficient conjugation, and sends 1 to 1. Inverses of unitary matrices are conjugate transposes, so (7.12) is a polynomial function of link entries and their conjugates. Substituting polynomials with zero constant term into a classical observable polynomial with $f_{d,i}$ factors leaves at least that many transported-curvature factors in every monomial. The final polynomial is not truncated again.

   For the scaling of jets in Definition 6.9, homogeneity in the transport equation and uniqueness give

   $$U_{J,\varepsilon} = U_{\mathrm{sc}_\varepsilon J,1}\,. \tag{7.13}$$

   The compositions $R_d^{[d]}\mathcal{Q}_d$ and $R_d\mathcal{Q}_d$ have equal Taylor coefficients through degree d at zero. Composing with $\mathrm{sc}_\varepsilon J$ and then v proves (6.21) for P_d^0.

2. *Restore the exact signed symmetries.* Average over the group of Definition 6.2:

   $$P_d[v](U) = \frac{1}{|H_4|}\sum_{h\in H_4} P_d^0[h^{-1}\cdot v](h^*U)\,. \tag{7.14}$$

   Changing the summation index gives the signed coordinate identity in (6.20); the unit and reality identities pass to the average as well. For the Taylor identity, the test links satisfy

   $$h^*U_{J,\varepsilon} = U_{T_{h^{-1}}J,\varepsilon} \qquad (h\in H_4)\,. \tag{7.15}$$

   This follows by substitution in (6.17) and uniqueness; on a reversed link the solution is $u(1)^{-1}u(1-t)$. In each summand of (7.14), the inverse action on polynomials from Definition 6.5 cancels the change in the test jet, preserving (6.21). A transformed transported plaquette is another transported plaquette, with a possible sign from reversal of its circulation. Since the action on polynomials preserves factor number, the lower factor bound survives averaging. Enclose all walks and plaquettes occurring in this finite sum in one centred integer cube Σ_d. This gives every condition in Definition 6.10.

3. *Bound derivatives on the good set.* For the bound, each transported curvature Q has value at most t on the good set. All links occurring in its fixed walks and plaquettes have maximum distance at most a fixed R from the origin on $\mathbb{Z}^4$. After periodic placement their distance is still at most R, so the allowed direction there has size at most $t(1+R)^q$. Each differentiation of a link product introduces one such direction. For $1\le r\le k$, this gives

   $$|Q(U)|_{\mathfrak{g}} \le t, \qquad |D_{A_1}\cdots D_{A_r}Q(U)|_{\mathfrak{g}} \le C_{r,q}t^r \le C_{r,q}t\,. \tag{7.16}$$

   The constant includes the fixed matrix projection and finitely many path factors. Leibniz's rule therefore gives Ct^s for every derivative through order k of a monomial with s factors. Sum its finitely many coefficients and use $s\ge f_{d,i}$ and $t\le 1$ to obtain (7.11). Repeated links on a small torus affect only the finite differentiation multiplicities, so the constant is independent of ℓ.

□

These polynomial functions are smooth along the link variations (7.9) at every configuration; the estimate applies on the closed good set, including its boundary. The derivatives are taken on the polynomial itself, before multiplying by any indicator of the good set. This distinction will also be used for the regular activities below.

7.3 Products of insertions

To carry one insertion through an identity of weights, introduce a commuting formal variable ζ with $\zeta^2 = 0$. The coefficient of ζ then records exactly one insertion. Several such variables record finite lists of insertions, and restrictions on allowed lists can be imposed in the same algebra.

Let V be a finite set indexing the formal variables, and let $\mathcal{R} \subseteq 2^V$ be the collection of allowed sets of insertions. Assume that $\mathcal{R}$ is closed under taking subsets and contains $\varnothing$ and every singleton. For $I \in \mathcal{R}$, write $\zeta_I = \prod_{v\in I} \zeta_v$. Define the algebra and its weighted coefficient norm by

$$\begin{aligned} \mathcal{A}_{\mathcal{R}} &= \bigoplus_{I\in\mathcal{R}} \mathbb{C}\zeta_I, \qquad a = \sum_{I\in\mathcal{R}} a_I \zeta_I, \quad 1 = \zeta_\varnothing, \\ \zeta_I\zeta_J &= \begin{cases} \zeta_{I\cup J}, & I\cap J = \varnothing,\ I\cup J \in \mathcal{R}, \\ 0, & \text{otherwise}, \end{cases} \\ \|a\|_\omega &= \sum_{I\in\mathcal{R}} \omega^I |a_I|, \qquad \omega^I = \prod_{v\in I} \omega_v, \quad \omega_v > 0\,. \end{aligned} \tag{7.17}$$

The variables commute; their squares, and products excluded by $\mathcal{R}$, are zero. Associativity follows by viewing this as the quotient of $\mathbb{C}[\zeta_v : v \in V]$ by those monomials.

The weights multiply exactly when disjoint insertions are combined. Thus the estimates below have no extra factor depending on the number of formal variables.

Proposition 7.3 (Square-zero products)**.** *For $a, b \in \mathcal{A}_{\mathcal{R}}$ of (7.17),*

$$\|1\|_\omega = 1, \qquad \|ab\|_\omega \le \|a\|_\omega \|b\|_\omega, \qquad \left\| \sum_{r=0}^{|V|} \frac{a^r}{r!} \right\|_\omega \le e^{\|a\|_\omega}\,. \tag{7.18}$$

If $a_\varnothing = b_\varnothing = 0$, then $a^{|V|+1} = 0$ and the finite exponential $\operatorname{Exp}(a) = \sum_{r=0}^{|V|} \frac{a^r}{r!}$ satisfies

$$\operatorname{Exp}(a+b) = \operatorname{Exp}(a)\operatorname{Exp}(b), \qquad \operatorname{Exp}(-a)\operatorname{Exp}(a) = 1\,. \tag{7.19}$$

Proof. The norm estimate follows because only disjoint variable sets multiply nontrivially, and their weights multiply. The exponential identities follow from nilpotence and the finite binomial formula. We prove these as follows.

1. *Product bound.* A nonzero coefficient in ab comes from disjoint sets I, J, with $\omega^{I\cup J} = \omega^I\omega^J$; hence
$$\|ab\|_\omega \le \sum_{\substack{I,J\in\mathcal{R} \\ I\cap J=\varnothing,\ I\cup J\in\mathcal{R}}} \omega^I\omega^J |a_I||b_J| \le \left(\sum_I \omega^I |a_I|\right)\left(\sum_J \omega^J |b_J|\right)\,.$$
2. *Norm of the finite exponential.* Since $\|1\|_\omega = 1$, submultiplicativity and the triangle inequality give
$$\|\operatorname{Exp}(a)\|_\omega \le \sum_{r=0}^{|V|} \frac{\|a\|_\omega^r}{r!} \le e^{\|a\|_\omega}.$$

3. *Exponential identities.* If the constant coefficient vanishes, every factor of a uses a nonempty set of variables. A product of more than $|V|$ such factors is zero. Thus $a^{|V|+1} = b^{|V|+1} = 0$, and the binomial formula yields

$$\operatorname{Exp}(a+b) = \operatorname{Exp}(a)\operatorname{Exp}(b), \qquad \operatorname{Exp}(-a)\operatorname{Exp}(a) = 1.$$

□

The same norm justifies countable summation: if $\|a_j\|_\omega \le c_j$ and $\sum_j c_j < \infty$, the partial sums are Cauchy and $\|\sum_j a_j\|_\omega \le \sum_j c_j$. Completeness follows from finite dimensionality. For one variable, let ρ be a scalar weight and I an observable, viewed as functions of the link configuration. For a finite index set J, let K_j be scalar factors and H_j the factors with an insertion. The insertion rule is then

$$[\zeta]\,\rho(1+\zeta I) = \rho I, \qquad [\zeta]\prod_{j\in J}(K_j+\zeta H_j) = \sum_{j\in J} H_j \prod_{i\in J\setminus\{j\}} K_i\,. \tag{7.20}$$

Here $[\zeta]$ means the coefficient of ζ. These identities explain why an inserted observable must be integrated together with its weight.

7.4 Local interactions

After integration over a scale, the effective weight is a sum of interactions on finite regions. Where all nearby plaquettes are small, these interactions can be controlled by derivatives. The remaining terms must keep track of the actual sites incident to large plaquettes. They are not discarded. An activity is the scalar function assigned to one such region; different indices may refer to the same region.

Use the lattice of Definition 2.2, the threshold $0 < t \le 1$, the derivative order $k \ge 6$ and distance exponent $q \ge 2$ from (7.10). The support weight will measure both the size of a region and the cost of joining its separated pieces.

Definition 7.4 (Connected ball covers)**.** Fix an integer radius $B \ge 1$ and $\alpha > 0$. For $a \in \Lambda_\ell$, let $\mathbb{B}_{\ell,B}(a)$ be its closed ball in the distance (7.7). A set of sites is connected if any two of its sites can be joined within the set by a finite sequence of steps of maximum distance at most one. For $X \subseteq \Lambda_\ell$, define

$$\begin{aligned} \mathbb{B}_{\ell,B}(a) &= \{x \in \Lambda_\ell : d_\ell(x,a) \le B\},\\ \mathfrak{c}_B(X) &= \min_{\substack{D\subseteq\Lambda_\ell\\ X\subseteq\bigcup_{a\in D}\mathbb{B}_{\ell,B}(a)\\ \bigcup_{a\in D}\mathbb{B}_{\ell,B}(a)\ \text{connected}}} |D|\,. \end{aligned} \tag{7.21}$$

Thus $\mathfrak{c}_B(\varnothing) = 0$; the support weight is $e^{\alpha \mathfrak{c}_B(X)}$.

The minimum exists: taking every site as a centre covers the torus, which is connected by coordinate steps.

We call the activities controlled by derivatives on good configurations the regular terms. Fix finite sets of sites $X_i \subseteq \Lambda_\ell$, sites $o_i \in X_i$, and real Borel gauge-invariant functions E_i supported on the links inside X_i in Definition 2.6. They have the derivatives measured by (7.9) on the good set $\mathcal{G}_t(X_i)$ of (7.8). Their contribution is $\mathbf{1}_{\mathcal{G}_t(X_i)}E_i$. For the other terms, fix finite sets of sites $Y_j \subseteq \Lambda_\ell$, nonempty sets $I_j \subseteq Y_j$, and real Borel gauge-invariant functions K_j such that

$$K_j(U) \ne 0 \implies Y_j \cap \mathcal{B}_t(U) = I_j \quad \text{and} \quad U \in \mathcal{E}_j\,. \tag{7.22}$$

The optional extra condition $\mathcal{E}_j$ is Borel and gauge invariant. Both K_j and $\mathcal{E}_j$ depend only on the internal links of Y_j and the bad-site indicators there. Thus I_j specifies the exact bad sites in that region, not just a subset of them. The fixed sets and extra conditions are retained when an insertion changes the coefficients.

Evaluating the bad-site indicators of (7.8), either for a regular term or for the conditions (7.22), may require extra links. For a set of sites Y, write vertices(p) for the four corner sites of a plaquette p. All such links have endpoints in

$$\widehat{Y} = Y \cup \bigcup_{\text{vertices}(p)\cap Y\neq\varnothing} \text{vertices}(p)\,. \tag{7.23}$$

Two of the regions Y_j may occur together only when they are disjoint; this condition uses Y_j, while $\widehat{Y}_j$ contains the endpoints of the links used to evaluate its indicators.

The two activity bounds count terms differently: regular terms are summed at their chosen site, and the other terms at every site in their support. Precisely, bounds $\delta, w \geq 0$ mean that there exist nonnegative majorants e_i, k_j (upper bounds for the individual activities) with

$$\begin{aligned} &\|E_i\|_{t,k,q;o_i,X_i} \leq e_i, \qquad \sup_U |K_j(U)| \leq k_j, \\ &\sum_{i:o_i=x} e^{\alpha \mathfrak{c}_B(X_i)} e_i \leq \delta, \qquad \sum_{j:x\in Y_j} e^{\alpha \mathfrak{c}_B(Y_j)} k_j \leq w \quad (x \in \Lambda_\ell)\,. \end{aligned} \tag{7.24}$$

The last two sums are convergent series of nonnegative terms. Indices range over $\mathbb{N}$ and zero functions may be added.

Let $c \in \mathbb{R}$, and let S be the part of the action kept separate from the activities, assumed bounded, real, Borel and gauge invariant; write $\|S\|_\infty = \sup_U |S(U)|$. For a finite selection $J \subset \mathbb{N}$ of indices, multiply the factors K_j. The sum $\mathcal{Z}_K$ retains only disjoint regions whose specified bad sites cover the entire measured bad set; E_t sums the regular contributions:

$$\begin{aligned} E_t(U) &= \sum_i \mathbf{1}_{\mathcal{G}_t(X_i)}(U) E_i(U), \\ \mathcal{Z}_K(U) &= \sum_{\substack{J\subset\mathbb{N} \text{ finite} \\ Y_j\ (j\in J) \text{ pairwise disjoint} \\ \bigcup_{j\in J} I_j = \mathcal{B}_t(U)}} \prod_{j\in J} K_j(U), \\ \rho(U) &= \exp\bigl(c|\Lambda_\ell| - S(U) - E_t(U)\bigr)\mathcal{Z}_K(U)\,. \end{aligned} \tag{7.25}$$

The empty selection contributes 1 precisely when $\mathcal{B}_t(U) = \varnothing$.

These formulas are useful only if their infinite sums converge absolutely. The per-site bounds give this convergence at each finite volume; identifying the represented weight with an integrated Wilson weight is a later step.

Proposition 7.5 (Convergence of the activity sums)**.** *Under* (7.22)–(7.24)*, both series in* (7.25) *converge absolutely and uniformly in* U*. They are Borel and gauge invariant, and*

$$\begin{aligned} &|E_t(U)| \leq |\Lambda_\ell|\delta, \qquad |\mathcal{Z}_K(U)| \leq e^{|\Lambda_\ell| w}, \\ &|\rho(U)| \leq \exp\bigl(c|\Lambda_\ell| + \|S\|_\infty + |\Lambda_\ell|(\delta + w)\bigr)\,. \end{aligned} \tag{7.26}$$

In particular ρ *is integrable against product Haar. If* $\mathcal{B}_t(U) = \varnothing$*, then* $\mathcal{Z}_K(U) = 1$*.*

Proof. At fixed finite volume, the per-site hypotheses give summable bounds for individual terms; the sum over compatible selections is then bounded by an unrestricted product. The argument has three steps.

1. *Sum individual activities.* Since each regular term has one chosen site and every Y_j is nonempty, the weights in (7.24) are at least one and give

$$\sum_i e_i \le |\Lambda_\ell|\delta, \qquad \sum_j k_j \le |\Lambda_\ell| w\,. \tag{7.27}$$

   The regular term $\mathbf{1}_{\mathcal{G}_t(X_i)} E_i$ is bounded by e_i everywhere, so its series converges uniformly and absolutely.

2. *Sum the selections.* The disjoint-region sum is dominated by all finite selections, whose configuration-independent majorant satisfies

$$\sum_{J\subset\mathbb{N}\ \text{finite}} \prod_{j\in J} k_j = \prod_{j\ge 0}(1+k_j) \le \exp\left(\sum_j k_j\right). \tag{7.28}$$

   The equality follows by expanding finite products and taking increasing limits; the inequality uses $1+s\le e^s$ for $s\ge 0$. This proves absolute uniform convergence and the first two bounds in (7.26).

3. *Pass the properties to the sum.* All indicators in (7.25) are Borel and gauge invariant, as are the activities, so the limits have these properties. The exponent in ρ is bounded using $|S(U)|\le\|S\|_\infty$ and $|E_t(U)|\le|\Lambda_\ell|\delta$; thus

$$|\rho(U)| \le e^{c|\Lambda_\ell|+\|S\|_\infty+|\Lambda_\ell|(\delta+w)}.$$

   Product Haar is a probability measure, so this bound gives integrability. If there are no bad sites, a nonempty selection cannot have empty union of its nonempty I_j; only the empty selection remains.

□

The estimates here allow the local bounds δ, w to be independent of volume, while their global sums grow with $|\Lambda_\ell|$. They justify summation and integration of supplied activities. The integration over a scale and subsequent regrouping must still construct those activities, impose the required conditions on their values and quadratic Taylor coefficients at $U(e)=1$, and prove the bounds for the same chosen family.

8 Infinite volume

We now pass from correlations in finite boxes to correlations on $\mathbb{R}^4$. The moments are those of Definition 2.9, for the one Wilson family of Proposition 5.1. The difficulty is to control a fixed collection of local observations as the surrounding box grows. The reflection inequality, which pairs an observation with its reflected conjugate, compares their joint expectation with repeated copies of each observation, whose bounds do not depend on box size.

The section follows five steps. Subsection 8.1 proves these reflection estimates. Subsection 8.2 uses them to bound changes in the insertion points and select common subsequences. Subsection 8.3

identifies the coefficients that disappear from all correlations and constructs fixed representatives of derivatives. Subsection 8.4 obtains smooth fields, their symmetries and reflection positivity. Finally, Subsection 8.5 uses the fixed-volume limit to remove the auxiliary spacing subsequence, leaving only a volume refinement.

The construction needs three inputs: the repeated-cell bounds (8.6), the polynomial comparisons of Subsection 8.3, and the fixed-volume convergence used in Subsection 8.5. They give correlations on $\mathbb{R}^4$; later, reflection positivity defines a scalar product and the rotation argument restores orthogonal covariance before the phase decomposition of Proposition 5.2.

8.1 Lattice estimates

The classical polynomials of Definition 6.4 provide the concrete observable indices of Definition 2.6: an index is a pair (d, v) with $v \in V_d$. Its insertion is the function (6.23). We retain the degree when following an insertion through the limiting procedure, since the same polynomial may belong to several V_d.

Definition 8.1 (Coefficient conventions)**.** For the Wilson measures of Definitions 2.7 and 2.4, write

$$\langle F\rangle_{N,p} = \int F \, d\mu_{N,p}, \qquad \langle F\rangle_{\beta} = \int F \, d\mu_{\ell,\beta} \tag{8.1}$$

when the current torus has ℓ sites per side. Choose norms on V_d using the fixed bases of Lemma 6.6, and equip linear maps with the induced operator norms. For the signed permutation group H_4 of Definition 6.2, write

$$\rho_d(h)v = h \cdot v, \qquad u_d = a^d_{r(d)}1, \qquad v \in V_d,\ h \in H_4.$$

The action is (6.12); $r(d)$ is the reference level of Definition 6.11, and u_d is the physical unit polynomial of Proposition 6.12. A list $v = (v_1, \dots, v_n)$ includes specified degrees d_i with $v_i \in V_{d_i}$. We suppress the pairs (d_i, v_i) in insertion and moment notation. For a coefficient or field list, $v_{i\to w}$ means replacement of its ith entry by w. The symbol $\partial_{i,\mu}$ means differentiation in coordinate μ of the ith physical insertion point.

On physical coordinate spaces, $|\cdot|$ denotes the Euclidean norm and $|\cdot|_\infty$ the maximum of the absolute coordinate values. The sites and links are those of Definition 2.2, with physical positions as in Definition 2.9.

The block factors ℓ_j are the odd integers relating successive spacings; the following conditions are specific to the constructed family, beyond Definition 2.7:

$$a_0 = 1, \qquad a_{j+1} = \frac{a_j}{\ell_j}, \qquad \ell_j \in 2\mathbb{N}+1, \quad \ell_j \geq 3, \qquad L_p \in 4\mathbb{N}, \qquad \beta_{N,p} \geq 0\,. \tag{8.2}$$

Thus one step at scale a_j comprises ℓ_j steps at scale a_{j+1}. We also require these block factors to grow slowly relative to the inverse spacing:

$$\frac{\log \ell_j}{\log a_j^{-1}} \longrightarrow 0 \qquad (j \longrightarrow \infty)\,. \tag{8.3}$$

The quotient at $j = 0$ plays no role. The translation, conjugation, signed-permutation and unit identities in Proposition 6.12 are assumed beyond the reference levels.

Definition 8.2 (Lattice reflections and cells)**.** For a coordinate unit vector e_μ, reflection in the physical plane $x^\mu = c$ is

$$r_{\mu,c}x = x + 2(c - x^\mu)e_\mu, \qquad \begin{cases} c \in a_N\mathbb{Z}, & \text{site reflection,} \\ c \in a_N\big(\mathbb{Z} + \frac{1}{2}\big), & \text{bond reflection.} \end{cases} \tag{8.4}$$

The site plane passes through lattice points; the bond plane lies halfway between two neighbouring layers of sites and bisects their connecting links. On the torus, coordinates are reduced modulo L_p. The pullback r^*U on link configurations is the coordinate action of Definition 6.2; a reversed link uses its inverse matrix. Set $\Theta_r F = \overline{F \circ r^*}$.

A cell tiling $\mathcal{C}$ divides the torus into congruent cubes, with the same even number of cells along each axis. All its boundary planes are site planes, or all are bond planes. An observable is interior to a cell or half-torus if it is bounded, Borel measurable, gauge invariant, and depends only on links whose two endpoints lie strictly inside that region. Choose one cell as the base cell, where the observable is initially given. For an observable F in that cell, let $\Theta_C F$ be its reflected copy in cell C, conjugated after an odd number of reflections. Its chessboard root is

$$\|F\|_{\mathcal{C}} = \left| \left\langle \prod_{C \in \mathcal{C}} \Theta_C F \right\rangle_{N,p} \right|^{\frac{1}{|\mathcal{C}|}}. \tag{8.5}$$

The centred tiling $\mathcal{C}_j(y)$ has cell side a_j and base cell centred at the physical point $\widehat{y}$ of Definition 2.9:

$$\big(\mathcal{C}_j(y)\big)_0 = \widehat{y} + \Big[-\frac{a_j}{2}, \frac{a_j}{2}\Big]^4 \pmod{L_p}.$$

Its boundary planes are bond planes because $\frac{a_j}{a_N}$ is odd for $N \geq j$.

This root compares a single local observation with copies filling the torus. The ultraviolet estimate bounds this quantity at every fixed physical cell size. For every $d \in \mathbb{N}$ and $R, \varepsilon > 0$, there exists $A \geq 0$ such that, for every $j \geq r(d)$, eventually in N, uniformly in p, y and $\|v\|_{V_d} \leq R$, the insertion is interior and

$$\|I_{N,p}(v, y)\|_{\mathcal{C}_j(y)} \leq A a_j^{-d-\varepsilon}. \tag{8.6}$$

The cutoff threshold may depend on j. A fixed finite collection of transformed polynomials is included by enlarging R.[7]

Unresolved. The ultraviolet part must establish (8.2), (8.3), (8.6) and the insertion identities for the particular family of Proposition 5.1. A bound for a different family would not suffice. These are the hypotheses used below, not consequences of Definition 2.8.[8]

Definition 8.3 (Ordered tails)**.** For chosen spacing and volume refinements N_j, p_k, an assertion $B(N, p)$ holds eventually on ordered tails if

$$\exists k_0\ \forall k \geq k_0\ \exists j_0(k)\ \forall j \geq j_0(k), \qquad B(N_j, p_k). \tag{8.7}$$

A coefficient or error tends to zero on these tails if its norm is eventually less than every positive tolerance in this order. Before a refinement is selected, use $N_j = j$ and $p_k = k$. A refining ultrafilter is a maximal proper filter on index pairs containing every such tail; it provides common limits for bounded families in compact finite-dimensional sets.

[7] Formal correspondence at revision `3456e5e`: `Extraction/Center.lean`: the uniform centred estimate and its projection to each extraction.

[8] Formal correspondence at revision `3456e5e`: `MergeProof/Gardens.lean` and `Source/Construction.lean`.

Thus the spacing threshold may depend on volume, in the order specified by Definition 2.12.

Reflection positivity pairs an observable with its reflected complex conjugate nonnegatively. The kernel form used to prove it will also control the movement of insertion points.

Definition 8.4 (Positive kernels)**.** A complex kernel K on a set E is positive if, for every finite list $X_i \in E$ and $c_i \in \mathbb{C}$,

$$\sum_{i,j} \bar{c}_i c_j K(X_i, X_j) \geq 0.$$

For a measured configuration space (E, μ), it acts by $(Kf)(X) = \int_E K(X,Y) f(Y)\, d\mu(Y)$ whenever the integral exists.

Lemma 8.5 (Reflection estimates)**.** *For the Wilson measure of Definition 2.4 on an even torus, $\beta \geq 0$, and bounded Borel gauge-invariant observables depending only on links whose endpoints lie strictly inside the indicated half-torus or cells,*

$$\langle \Theta_r F\, F \rangle_\beta \geq 0 \qquad \textit{(r a site or bond reflection)}, \tag{8.8}$$

$$\left| \left\langle \prod_{C \in \mathcal{C}} \Theta_C F_C \right\rangle_\beta \right| \leq \prod_{C \in \mathcal{C}} \|F_C\|_{\mathcal{C}}, \tag{8.9}$$

$$\|F + G\|_{\mathcal{C}} \leq \|F\|_{\mathcal{C}} + \|G\|_{\mathcal{C}}.$$

The reflection planes and chessboard roots are those of Definition 8.2; the observables depend only on links whose endpoints lie strictly inside the stated regions.

Proof. The proof has three steps.

Step 1. The boundary kernel is positive, so integration over boundary data gives $\langle \Theta_r F\, F \rangle_\beta \geq 0$.

Step 2. Reflection positivity gives the fold inequality $\mathcal{Z}(i)^2 \leq \mathcal{Z}(i^+)\mathcal{Z}(i^-)$; repeated folds make a maximising assignment constant.

Step 3. Divide each cell observable by its chessboard root plus η, apply the constant-assignment bound, and let $\eta \downarrow 0$.

Positivity. For the reflection plane under consideration, call an unoriented link $\{x, x + a_N e_\mu\}$ a crossing link when its endpoints lie on opposite sides of the plane. The real inner product $\operatorname{Re}\operatorname{tr}(UV^*)$ is a positive kernel on G in Definition 8.4. Entrywise products of positive matrices and the exponential series preserve positivity, so the Wilson interaction with crossing links set to the identity, including half of each boundary slice's weight on either side, gives a continuous positive kernel $K_0(X, Y)$ of the two slice configurations X, Y. Vertexwise gauge transformations on a slice are written $g \cdot X$, as in Definition 2.5. Haar invariance restores the crossing links by averaging these transformations independently:

$$K(X,Y) = \iint K_0(g \cdot X, h \cdot Y)\, dg\, dh.$$

To see positivity after averaging, let μ_X be the distribution of $g \cdot X$ under Haar measure and put $\nu = \sum_i c_i \mu_{X_i}$. Then

$$\sum_{i,j} \bar{c}_i c_j K(X_i, X_j) = \iint K_0(X,Y)\, d\bar{\nu}(X)\, d\nu(Y) \geq 0. \tag{8.10}$$

The last inequality follows by approximation with finite sums in the positive quadratic form of K_0. The integrand with independent g, h need not be nonnegative.

Now split the lattice into fixed boundary link data u and open-half configurations v, w. The boundary kernel after integrating crossing links is

$$P_u(v,w) = \begin{cases} 1, & \text{site reflection,} \\ K\big(X_-(w), X_-(v)\big) K\big(X_+(v), X_+(w)\big), & \text{bond reflection,} \end{cases}$$

where X_-, X_+ read the first and last slice of a half-torus. It is positive: in the bond case it is a product of two positive kernels, one with exchanged arguments. Let Z be the Wilson normalising constant of (2.4), and $A_u(v)$ the half-weight in this factorisation of its density. Let $U_{u,v}$ be any full configuration agreeing with u, v on the boundary and chosen half. Set $f_u(v) = A_u(v)F(U_{u,v})$. Because the observable uses only links whose endpoints lie in the chosen half, its value is independent of unused links in a completion to the full torus. With normalised product Haar measures on these data, the exact factorisation is

$$\langle \Theta_r F\, F \rangle_\beta = \frac{1}{Z} \int \left[\iint P_u(v,w) f_u(v) \overline{f_u(w)}\, dv\, dw \right] du \geq 0. \tag{8.11}$$

All functions are measurable and bounded on compact configuration spaces, so the changes of integration order are valid.[9]

Repeated reflections. Choose a finite list of base-cell observables $F_1, \dots, F_s$. An assignment $i = (i_C)_{C \in \mathcal{C}}$ chooses which observable occupies each cell, and we write

$$\mathcal{Z}(i) = \left| \left\langle \prod_{C \in \mathcal{C}} \Theta_C F_{i_C} \right\rangle_\beta \right|.$$

A fold keeps the assignment on one half-torus and reflects it onto the other half. Let i^+, i^- be the two choices. Cauchy–Schwarz for the reflected form gives

$$\mathcal{Z}(i)^2 \leq \mathcal{Z}(i^+)\mathcal{Z}(i^-). \tag{8.12}$$

Thus a maximising assignment stays maximising under either fold:

$$M = \max_i \mathcal{Z}(i), \quad \mathcal{Z}(i) = M \quad \Longrightarrow \quad M^2 \leq \mathcal{Z}(i^+)\mathcal{Z}(i^-) \leq M^2.$$

If $M = 0$, every assignment is maximising; otherwise both factors equal M.

It remains to make a maximising assignment constant. Fix a coordinate with $2q$ cells, and call a band of width w constant in this coordinate when the assignment does not change along the coordinate inside that band. Every assignment has such a band of width one. A fold at its initial boundary copies the first layer to the preceding layer; it preserves all the existing layers. With t denoting the band's initial cell, the two useful steps are

$$(t, w) \longmapsto (t-1, w+1) \quad (1 \leq w < q), \qquad (t, q) \longmapsto \text{the whole coordinate circle.}$$

The second step reflects a constant half-circle onto its complement. These folds preserve constancy in coordinates already treated. Doing this in all four coordinates gives a constant maximising assignment.

[9] Formal correspondence at revision `3456e5e`: `YM/Slice/Positive.lean`, `averaged` and `step_positive`; `YM/Slice/Gate.lean`, `wall`, `links` and `portPos`; `YM/Slice/Reflection.lean`, `readers` and `link_positive`: the exact boundary form.

Normalisation. For the observables in (8.9), set

$$\widetilde{F}_C = \frac{F_C}{\|F_C\|_{\mathcal{C}} + \eta}, \qquad \eta > 0.$$

Every constant assignment from this finite list has expectation of absolute value at most 1, by (8.5). The maximising argument therefore gives

$$\left|\left\langle \prod_C \Theta_C \widetilde{F}_C \right\rangle_\beta\right| \le 1, \qquad \left|\left\langle \prod_C \Theta_C F_C \right\rangle_\beta\right| \le \prod_C \bigl(\|F_C\|_{\mathcal{C}} + \eta\bigr).$$

Letting $\eta \downarrow 0$ proves (8.9). Finally, expand the reflected product of $F + G$ and apply that estimate term by term:

$$\|F + G\|_{\mathcal{C}}^{|\mathcal{C}|} \le \bigl(\|F\|_{\mathcal{C}} + \|G\|_{\mathcal{C}}\bigr)^{|\mathcal{C}|}.$$

Taking the root proves subadditivity.[10] □

8.2 Bounds and compactness

The compactness argument has three steps.

Step 1. Choose cell sizes with $a_m < r$ and $6na_k < a_m$; the centred root bound then gives a uniform bound when the insertion points are pairwise at least r apart.

Step 2. Moving one insertion by h lattice steps changes adjacent slice gaps from (Y_i, Y_j) to $(Y_i + h, Y_j - h)$; positivity bounds the resulting correlation difference by a constant times the physical displacement.

Step 3. Local boundedness and this movement estimate give uniform Cauchy convergence on compact sets; a diagonal choice over the countable polynomial bases and compact exhaustions gives one subsequence for all correlations.

We first separate points by planes perpendicular to each coordinate axis. For example, if their first coordinates are distinct, a plane perpendicular to the first axis can be placed between any two consecutive first coordinates. The empty interval around each such plane is where an insertion can move without reaching another insertion. This explains the temporary restriction below.

Definition 8.6 (Coordinate separation)**.** For the configuration space of Definition 2.10, put

$$\mathrm{Gen}_n = \{x \in \mathrm{Conf}_n : x_i^\mu \ne x_j^\mu \text{ for } i \ne j,\ \mu = 1, \dots, 4\}. \tag{8.13}$$

At fixed period L_p, let $\mathrm{Gen}_n^{(p)}$ be the same set with inequalities taken modulo $L_p\mathbb{Z}$. These are the coordinate-separated configurations, in ordinary space and on the periodic lattice respectively.

Both sets are open, and their complements in the corresponding distinct-point configuration spaces have measure zero. The periodic set lies in the periodically separated set of Definition 2.13. Two distinct points with the same first coordinate are excluded for now; Subsection 8.4 will recover their correlations by smoothness.

To compare different spacings on a common domain, extend each lattice correlation to a step function on physical space.

[10] Formal correspondence at revision `3456e5e`: `Cells/Flood.lean`: extension of a band by one cell and successive coordinates; `Cells/Chess.lean`: maximisation, normalisation and subadditivity.

Definition 8.7 (Lattice extensions)**.** With $\ell_{N,p} = \frac{L_p}{a_N}$ as in (2.7), round each coordinate up to the next lattice point and reduce it modulo the period:

$$y_{N,p}(x) = \big[\lceil a_N^{-1} x \rceil\big] \in \Lambda^n_{\ell_{N,p}}, \qquad M^{\mathrm{step}}_{N,p}(v;x) = M_{N,p}(v; y_{N,p}(x)). \tag{8.14}$$

Brackets denote the periodic class, the ceiling acts componentwise, and $M_{N,p}$ is the moment of Definition 2.9. The physical displacement caused by rounding is at most $4na_N$ modulo L_p.

Proposition 8.8 (Auxiliary extraction)**.** *Assume* (8.2), (8.3) *and* (8.6) *and the insertion identities of Proposition 6.12. Every pair of strictly increasing spacing and volume sequences has refinements N_j, p_k and multilinear kernels P_p, S_0 with the following properties, simultaneously for all $n \in \mathbb{N}$, degrees $d_i \in \mathbb{N}$ and coefficients $v_i \in V_{d_i}$.*

(i) *Locally uniformly on the sets of Definition 8.6,*

$$\begin{aligned} M^{\mathrm{step}}_{N_j,p}(v;\cdot) &\longrightarrow P_p(v;\cdot) \in C(\mathrm{Gen}_n^{(p)}) \quad (p \in \mathbb{N}), \\ P_{p_k}(v;\cdot) &\longrightarrow S_0(v;\cdot) \in C(\mathrm{Gen}_n). \end{aligned} \tag{8.15}$$

(ii) *Every separated test f of Definition 2.10 is eventually supported inside volume p_k (Definition 2.11), has $fS_0(v;\cdot) \in L^1$, and admits numbers $R_p(v;f)$ such that*

$$\begin{aligned} &\lim_{j\to\infty} C_{N_j,p}(v;f) = R_p(v;f) \qquad\qquad (f \text{ supported inside volume } p), \\ &\lim_{k\to\infty} R_{p_k}(v;f) = \int f(x) S_0(v;x)\,dx. \end{aligned} \tag{8.16}$$

Proof. The extraction has four steps.

Step 1. The centred roots give local bounds on separated configurations; the cell choices satisfy $a_m < r$ and $6na_k < a_m$.

Step 2. Slice positivity controls movements by $|M^{\mathrm{step}}_{N,p}(v;x) - M^{\mathrm{step}}_{N,p}(v;y)| \le C_K|x-y| + o(1)$.

Step 3. A finite-net argument gives locally uniform limits, first at each volume and then along one common volume refinement.

Step 4. Dominated convergence of the step-function smearings gives the two ordered limits for every separated test.

Local bounds. The centred estimate (8.6) controls an observable repeated in cells centred on its insertion site. It also controls repetitions in translated cells, provided the insertion stays away from their boundaries. At aligned smaller-cell centres, apply (8.9) and put 1 in the unused cells. Between these centres, let $R(z)$ be the chessboard root as the insertion moves by z lattice steps along one axis. A fold gives the discrete convexity inequality

$$R(z)^2 \le R(z-1)R(z+1) \quad \Longrightarrow \quad R(z) \le \max\{R(z_-), R(z_+)\} \qquad (z_- \le z \le z_+).$$

Here z_-, z_+ are consecutive aligned centres. Applying this in each direction bounds every translated root by the largest centred root of the reflected coefficients. It suffices that the insertion site is at least $3a_k$ from each boundary and its supporting links lie within distance a_k of that site.[11]

Now suppose n insertion sites have pairwise physical torus sup-distance at least $r > 0$. Choose a cell size $a_m < r$ to put different sites in different cells, and then a smaller size a_k for the boundary

[11] Formal correspondence at revision `3456e5e`: `YM/Extraction/Around.lean`: translated cells; `YM/Cells/Slide.lean`: discrete convexity.

margins. The forbidden translations in each coordinate have total length at most $6na_k$ in one period a_m. Consequently

$$m \le k, \quad a_m < r, \quad 6na_k < a_m \quad \Longrightarrow \quad \left| \left\langle \prod_{i=1}^{n} I_{N,p}(v_i, y_i) \right\rangle_{N,p} \right| \le A^n \tag{8.17}$$

for sufficiently large N, where A bounds the centred roots at level k for the finitely many reflected coefficients. Choose a translation outside the forbidden intervals and apply the translated-root bound and (8.9). The finite supports fit inside their cells once N is sufficiently large. Thus fixed separated compact sets have local bounds, uniform in volume on the ordered tails of Definition 8.3.

Moving an insertion. For at least two insertions, fix an axis and integrate the lattice one slice at a time, perpendicular to that axis. The positive integral kernel K propagates through one empty slice; B_i integrates a slab containing the ith insertion. Let Y_i count the empty slices before the next slab. The trace closes this chain around the periodic direction. If the torus has M slices and each slab has w slices, the Wilson factorisation gives

$$\langle \prod_i I_i \rangle = \frac{\operatorname{Tr}(B_1 K^{Y_1} \cdots B_n K^{Y_n})}{Z}, \qquad Z = \operatorname{Tr} K^M > 0, \qquad nw + \sum_i Y_i = M.$$

Here $I_i = I_{N,p}(v_i, y_i)$; the trace is the cyclic integral of these kernels over the slice configurations. The slice kernel and inserted kernels are continuous, since the original insertions are continuous by Reg2. The adjoint kernel is $B_i^*(X, Y) = \overline{B_i(Y, X)}$. Let A_i bound the chessboard root of the ith insertion. For $M = 2qw$, the alternating reflected product and (8.9) give

$$\left(\frac{\operatorname{Tr}\big((B_i^* B_i)^q\big)}{Z} \right)^{\frac{1}{2q}} \le A_i,$$

The period condition in (8.2) makes q an integer when the slabs have twice a centred cell's width.

Moving one slab by h steps changes the two adjacent distances by $+h$ and $-h$. Approximate the cyclic integrals by finite weighted sums with nonnegative weights. For the resulting positive matrix K, define $K^z = \exp(z \log K)$ on its positive eigenspaces and set it equal to zero on its kernel. Its matrix entries are entire in z, and it agrees with the ordinary powers at positive integer exponents. For the two adjacent interval indices i, j, set

$$T(z) = \operatorname{Tr}\big(B_1 K^{Y_1(z)} \cdots B_n K^{Y_n(z)}\big), \qquad Y_l(z) = \begin{cases} Y_i + z, & l = i, \\ Y_j - z, & l = j, \\ Y_l, & l \ne i, j. \end{cases}$$

For a finite matrix, let B^* be its conjugate transpose and $|B| = (B^*B)^{\frac{1}{2}}$. Its Schatten norm is $\|B\|_s = (\operatorname{Tr}|B|^s)^{\frac{1}{s}}$, the ℓ^s norm of its singular values. Choose $b > 0$ with all $Y_l > 2b$. For $|z| \le 2b$, imaginary powers have operator norm at most one, and Schatten Hölder gives

$$|T(z)| \le \prod_l \|B_l\|_{2q} \, Z^{1 - \frac{n}{2q}}, \qquad \frac{n}{2q} + \frac{\sum_l \operatorname{Re} Y_l(z)}{M} = 1.$$

For $0 \le h \le b$, Cauchy's estimate on discs of radius b centred on $[0, h]$ therefore yields

$$\frac{|T(h) - T(0)|}{Z} \le \frac{h}{b} \prod_l \left(\frac{\operatorname{Tr}|B_l|^{2q}}{Z} \right)^{\frac{1}{2q}} \le \frac{h}{b} \prod_l A_l.$$

Pass to the limit in the finite weighted sums. The integer endpoint exponents stay positive, so the choice on the zero eigenspace changes neither endpoint.[12]

On a compact coordinate-separated set, choose fixed slabs leaving a positive physical interval between successive insertions in each coordinate direction. The integer b can then be chosen with $a_N b$ bounded below, independently of sufficiently fine N. The preceding estimate costs at most a constant times the physical displacement $a_N h$. Move the $4n$ coordinates in turn and include the rounding error of Definition 8.7. On each compact neighbourhood K in $\mathrm{Gen}_n^{(p)}$, this gives, for sufficiently close $x, y \in K$,

$$|M_{N,p}^{\mathrm{step}}(v;x) - M_{N,p}^{\mathrm{step}}(v;y)| \le C_K|x-y| + o(1), \qquad N \longrightarrow \infty, \tag{8.18}$$

uniformly in x, y. For fixed $K \subset \mathrm{Gen}_n$, the same constants work independently of sufficiently large volumes on ordered tails. For one insertion use translation invariance; for no insertions the moment is 1.

Common subsequences. Fix a coefficient basis tuple and a compact set K. Local boundedness gives a subsequence converging at every point of a countable dense subset. Choose a finite δ-net E_δ from that subset, so every point of K is within δ of a point of E_δ. Then (8.18) gives

$$\sup_K |M_j - M_l| \le 2C_K\delta + o_j(1) + o_l(1) + \max_{z\in E_\delta} |M_j(z) - M_l(z)|.$$

Here M_j denotes the corresponding lattice extension along the selected subsequence. First let $j, l \to \infty$ and then $\delta \downarrow 0$. The sequence is uniformly Cauchy on K, and its limit is continuous. This is the compactness mechanism of Arzelà–Ascoli, even though the approximating step functions need not be continuous.

Take the diagonal over all fixed volumes, numbers of insertions, degree lists, basis tuples and compact exhaustions. These choices are countable, so one spacing subsequence gives every P_p in (8.15). Pass the volume-independent bounds and movement estimates to these limits. The same argument gives one volume subsequence and S_0. Extend from the bases by multilinearity. Set these raw kernels to zero off their coordinate-separated domains; their complements have measure zero.

Smeared limits. Write the weighted lattice sum as an integral of the associated step functions. A compact neighbourhood of $\operatorname{supp} f$ stays separated, and at a volume containing the support it avoids periodic collisions. The local bounds dominate the integrands there, while rounding changes f uniformly by $o(1)$. Dominated convergence first in N_j and then in p_k gives both limits in (8.16). Finally $L_{p_k} \to \infty$ ensures eventual containment of the support.[13] □

8.3 Equation relations

Compactness first gives correlations where every pair has distinct coordinate projections. Integration by parts represents their weak derivatives by higher-degree polynomials. The comparison inputs identify these polynomials modulo invisible coefficients, while the separate rank bound keeps their representatives bounded.

We now state these additional ultraviolet inputs precisely. Along a further common refinement, preserving Proposition 8.8, assume linear projections $P_{d,N,p}, P_d : V_d \to V_d$ with

$$P_{d,N,p}^2 = P_{d,N,p}, \qquad P_{d,N,p} \longrightarrow P_d, \qquad P_d^2 = P_d \quad \text{in operator norm.} \tag{8.19}$$

[12] Formal correspondence at revision `3456e5e`: `YM/Complex/Gaps.lean`: finite trace difference; `YM/Slice/Calendar.lean`: passage to kernels; `YM/Extraction/Difference.lean`: insertion movements.

[13] Formal correspondence at revision `3456e5e`: `YM/Extraction/Lipschitz.lean`: common subsequences; `YM/Extraction/Integrate.lean`: dominated convergence; `YM/Extraction/Held.lean`: their combination.

All limits and eventual assertions in this subsection use the ordered tails of Definition 8.3. The inverse coefficient transfer $\tau^{-1}_{N,p,d}$ of (6.24) measures the size of a coefficient family $w_{N,p}$ at the fine cutoff. We allow families satisfying

$$\forall \varepsilon > 0, \qquad \|\tau^{-1}_{N,p,d} w_{N,p}\|_{V_d} \le a_N^{-\varepsilon} \quad \text{eventually.} \tag{8.20}$$

The ultraviolet estimates must imply (8.20) for bounded coefficient families. They do so by bounding the growth at each intervening scale: the logarithm of the product of these bounds is $o(\log a_N^{-1})$.

Definition 8.9 (Cell errors)**.** An error $E_{N,p}(y)$ is a lattice observable, with its link configuration argument suppressed. It is small in reflected cells if one nonnegative rate $\eta_{N,p} \to 0$ along ordered tails has the following property. For every integer $q \ge 0$ and lower level m, there exists $k \ge m$ such that eventually, for every translated bond cell of side a_k and every insertion site with integer coordinates z measured from the cell's first site in each direction and satisfying

$$M \le z^\mu, \quad z^\mu + M + 1 \le \frac{a_k}{a_N}, \quad M = \left\lfloor \frac{a_k}{8(q+1)a_N} \right\rfloor \quad (\mu = 1, \dots, 4), \tag{8.21}$$

the error is a bounded Borel gauge-invariant observable depending only on links whose endpoints lie strictly inside that cell, as in Definition 8.2, and its chessboard root (8.5) is at most $\eta_{N,p}$. This condition keeps the error away from cell boundaries, where the chessboard estimate can be applied. A degree-e representative means an insertion $I_{N,p}(w_{N,p}, y)$ with $w_{N,p} \in V_e$.

At every degree we require the following three comparisons:

(C1) Each fixed degree-d insertion has a degree-e representative satisfying (8.20) for $d \le e$, differing by an error small in reflected cells.

(C2) The forward difference $\frac{I_{N,p}(v,y+e_\mu) - I_{N,p}(v,y)}{a_N}$ has such a representative of degree $d+1$.

(C3) A family satisfying (8.20) in $\ker P_{d,N,p}$ differs by such an error from bounded measurable observables whose correlations with every fixed list of the other insertions vanish exactly, eventually at each positive separation.

Boundedness in C3 is only at each cutoff and site, which suffices for integration against the finite Wilson law.

Fix a number of insertions, one slot i (one position in the list), all coefficients in the other slots, and a separated test f of Definition 2.10. Varying the coefficient $w \in V_d$ in that slot gives linear functionals

$$L_{N,p}(w) = C_{N,p}(v_{i\to w}; f), \qquad L(w) = \int f(x) S_0(v_{i\to w}; x)\, dx, \qquad N_d = \bigcap_L \ker L. \tag{8.22}$$

The intersection includes every choice of the number of insertions, position, other polynomials and test. Thus N_d consists of coefficients invisible to every separated smeared correlation. In the following rank input and comparisons, eventual assertions retain Definition 8.3:

$$\dim_{\mathbb{C}} N_d \le \dim_{\mathbb{C}} \ker P_d\,. \tag{8.23}$$

The projectors and this inequality are obtained on a further common refinement, retaining all previous limits.

Unresolved. Supply comparisons C1–C3, the coefficient bound (8.20) for bounded families, and the finite-dimensional detection argument for (8.23) for the chosen family. The dimension inequality is additional to the centred-root estimate.[14]

The following lemma replaces the coefficients depending on the fine cutoff in these comparisons by fixed coefficients. Its conclusion holds against every list of other fields, so that it gives genuine field operations rather than representatives depending on a particular test.

Lemma 8.10 (Fixed representatives)**.** *Assume the hypotheses of Proposition 8.8, the projections (8.19), the coefficient bound (8.20) for bounded families, the error condition (8.21), the rank bound (8.23), and comparisons C1–C3. Then $N_d = \ker P_d$, and there are complex-linear maps*

$$\kappa_{d,e} : V_d \longrightarrow V_e \quad (d \le e), \qquad \delta_{d,\mu} : V_d \longrightarrow V_{d+1}$$

such that, for every separated test of Definition 2.10 and functionals (8.22),

$$\begin{aligned} \int f S_0(v_{i\to\kappa_{d,e}v_i}) &= \int f S_0(v), \\ \int f S_0(v_{i\to\delta_{d,\mu}v_i}) &= -\int (\partial_{i,\mu} f) S_0(v). \end{aligned} \tag{8.24}$$

Proof. The proof has two steps.

Step 1. Identify the invisible directions using $\ker P_d \subset N_d$ and $\dim N_d \le \dim \ker P_d$, hence $N_d = \ker P_d$.

Step 2. Divide each projected representative by $1 + \|P_{e,N,p} w_{N,p}\|$; compactness gives a nonzero normalised limit, which can be rescaled to represent every target correlation at once.

An error small in reflected cells has vanishing separated smearings. Choose its cell below the fixed separation and a finer centred scale for the other insertions. Avoid boundary strips as before, with margin M; the factor $8(q+1)$ permits this for any fixed number of slots. The translated-cell estimate and (8.9) bound the mixed moment by $C\eta_{N,p}$. The weighted count of sites meeting $\operatorname{supp} f$ is bounded by the volume of a fixed neighbourhood, independently of N, p. Thus its smear tends to zero.

The maps $L_{N,p}$ converge in operator norm because they converge on a finite basis. For fixed v, the bounded family $v - P_{d,N,p} v$ belongs to the projection kernel. Comparison C3 gives $L(v - P_d v) = 0$ for every L. The dimension input closes the identification:

$$\ker P_d \subset N_d, \qquad \dim_{\mathbb{C}} N_d \le \dim_{\mathbb{C}} \ker P_d \quad \Longrightarrow \quad N_d = \ker P_d.$$

Let $\Delta_{i,\mu}$ be the forward lattice difference in slot i and direction μ, and let $e_{i,\mu}$ be that coordinate vector in the point list. Summation by parts on the torus gives

$$a_N^{4n} \sum_y f(\widehat{y}) \Delta_{i,\mu} M_{N,p}(v; y) = a_N^{4n} \sum_y \frac{f(\widehat{y} - a_N e_{i,\mu}) - f(\widehat{y})}{a_N} M_{N,p}(v; y)\,.$$

The test support stays inside the fundamental box eventually, so there is no wrapping. Its backward quotient tends uniformly to $-\partial_{i,\mu} f$ on a fixed neighbourhood. Local boundedness and the bounded weighted site count make the error vanish. Comparisons C1–C2 therefore give coefficient families $w_{N,p}$ with the required target limits for every choice of the other insertions and test at once.

[14] Formal correspondence at revision `3456e5e`: `MergeProof/Powder.lean` and `Next/RANK/Beam.lean`: same-family assembly and retained rank comparison.

Write $p_{N,p} = P_{e,N,p} w_{N,p}$, where e is the representing degree. Comparison C3 applied to the normalised remainder gives

$$L_{N,p}\left(\frac{w_{N,p} - p_{N,p}}{1 + \|p_{N,p}\|_{V_e}}\right) \longrightarrow 0\,.$$

Division by $1 + \|p_{N,p}\|_{V_e} \geq 1$ preserves (8.20), and the normalised projected vector is bounded. Take a cluster limit (c, z) of

$$\left(\frac{1}{1 + \|p_{N,p}\|_{V_e}}, \frac{p_{N,p}}{1 + \|p_{N,p}\|_{V_e}}\right)$$

in $[0,1] \times \overline{B}_{V_e}(0,1)$, where $\overline{B}_{V_e}(0,1)$ is the closed unit ball. Use an ultrafilter containing every ordered tail of Definition 8.3; this selects one common cluster limit for all finite-dimensional coordinates. If t_L is the target correlation or weak derivative in (8.24), the limits satisfy $P_e z = z$, $c + \|z\|_{V_e} = 1$ and $Lz = c\,t_L$. A vanishing c would contradict the normalisation:

$$c = 0 \implies z \in \operatorname{im} P_e \cap \ker P_e = \{0\} \implies c + \|z\|_{V_e} = 0 \neq 1.$$

Thus $\frac{z}{c}$ represents every target simultaneously. Choose representatives on each source basis and extend linearly. No further subsequence has been chosen for the uncountable collection of tests.[15] □

8.4 Smooth fields and positivity

Repeated use of Lemma 8.10 gives weak derivatives of every order, hence smooth representatives even where coordinate projections coincide. Quotienting coefficients invisible in every correlation gives the continuum field spaces; their degree maps and derivatives are independent of representatives and enter Definition 2.17.

Definition 8.11 (Continuum fields)**.** For the null subspace N_d in (8.22), set

$$F_d = V_d/N_d, \qquad \pi_d v = [v], \qquad \|[v]\|_{F_d} = \inf_{w \in N_d} \|v + w\|_{V_d}. \tag{8.25}$$

These are normed finite-dimensional spaces. The class $\pi_d u_d$ is denoted by $\mathbf{1}$ at degree d.

Definition 8.12 (Analytic conventions)**.** For a function on $\mathbb{R}^a$, D_x^s denotes its sth total Fréchet derivative, with its operator norm. For a multi-index $\alpha \in \mathbb{N}^a$, put $|\alpha| = \sum_l \alpha_l$ and $D^\alpha = \partial_1^{\alpha_1} \cdots \partial_a^{\alpha_a}$. For $u \in L^1(\mathbb{R}^a)$, use the Fourier transform

$$\widehat{u}(\xi) = \int_{\mathbb{R}^a} e^{-i\xi\cdot x} u(x)\, dx.$$

In applications to n insertion points, $a = 4n$.

Proposition 8.13 (Smooth correlations)**.** *Under the hypotheses of Lemma 8.10, there is a collection $S = (S_n)_{n \geq 0}$ of multilinear kernels on the polynomial lists of Definition 8.1, with quotient spaces as in Definition 8.11, such that:*

(i) *$S_n(v;\cdot) \in C^\infty(\mathrm{Conf}_n)$ on (2.11), $S_n|_{\mathrm{Gen}_n} = S_0$ on (8.13), and (8.16) holds with S_n.*

(ii) *$\ker \pi_d = N_d = \ker P_d$ is the null space of S: a coefficient lies in N_d exactly when every separated-point correlation containing it vanishes.*

[15] Formal correspondence at revision `3456e5e`: `Work/D01.lean`–`D04.lean`: coefficient costs, errors and differences; `Dash/Limits.lean`: normalised compactness; `Dash/Inert.lean`: fixed representatives.

(iii) *The maps $\kappa_{d,e}, \delta_{d,\mu}$ of Lemma 8.10 induce linear maps $j_{d,e} : F_d \hookrightarrow F_e$ and $\partial_\mu : F_d \to F_{d+1}$, with $j_{e,f} j_{d,e} = j_{d,f}$ and $j_{d,d} = 1$, satisfying*

$$\begin{aligned} S_n(\dots, j_{d,e} A_i, \dots; x) &= S_n(\dots, A_i, \dots; x), \\ S_n(\dots, \partial_\mu A_i, \dots; x) &= \partial_{i,\mu} S_n(\dots, A_i, \dots; x). \end{aligned} \tag{8.26}$$

The notation $S_n(A; x)$ denotes the induced correlation of quotient fields, with $A_i = \pi_{d_i} v_i$.

Proof. The proof has three steps.

Step 1. The weak identities (8.24) represent each coordinate derivative by another locally integrable correlation.

Step 2. For every compactly supported smooth cutoff χ, $\sup_\xi (1 + |\xi|)^b |\widehat{\chi S_0(v)}(\xi)| < \infty$ for all b, so Fourier inversion gives a smooth local representative.

Step 3. These representatives agree on overlaps; their pointwise identities descend to the quotient fields.

Equation (8.24) represents every weak coordinate derivative by another locally integrable kernel. Iteration gives derivatives of every order. For a smooth cutoff function χ supported in Conf_n, meaning a smooth function with compact support there, each derivative of $\chi S_0(v)$ is a finite sum of integrable compactly supported kernels. Integration by parts in the Fourier transform gives, for every integer $b \geq 0$,

$$\sup_\xi (1 + |\xi|)^b \left| \widehat{\chi S_0(v)}(\xi) \right| < \infty \,.$$

Fourier inversion supplies a smooth representative. Choose $\chi = 1$ near an arbitrary distinct-point configuration. The local representatives agree almost everywhere on overlaps, hence everywhere there by continuity. They glue to S and equal S_0 on the set (8.13). The same argument preserves multilinearity and smearings.[16]

The null-space description in 8.13(ii) follows in both directions: pointwise vanishing implies vanishing smearings, while smoothness and testing in small neighbourhoods recover pointwise vanishing from (8.22). The identities (8.24) then hold pointwise. A null input has a null image under κ or δ, since all its correlations, and hence their derivatives, vanish. Thus the maps descend to F_d. Applying the same identities to all other fields proves injectivity of $j_{d,e}$, its composition law and $j_{d,d} = 1$. □

The symmetries and positivity of the Wilson measure survive the ordered limit. The scale condition (8.3) also converts the bounds at fixed cell sizes into bounds in terms of the distance between insertions. We spell these out because they are the inputs to the later Hilbert-space construction.

For $r_\mu = r_{\mu,0}$ of Definition 8.2, a positive configuration has $x_i^\mu > 0$ and pairwise distinct points. For two such lists of lengths n and m, define the reflected correlation

$$G_S\big((v, x), (w, y)\big) = S_{n+m}(\rho(r_\mu)\bar{v}, w; r_\mu x, y). \tag{8.27}$$

Reflection acts on every point and coefficient in the first list. For joint separated tests f_α of Definition 2.10 supported on positive configurations, put

$$G_{\alpha\beta}(f) = \iint \overline{f_\alpha(x)} f_\beta(y) G_S\big((v^\alpha, x), (v^\beta, y)\big)\, dx\, dy. \tag{8.28}$$

[16] Formal correspondence at revision `3456e5e`: `Dash/Regular.lean`, `Dash/Twist.lean`, and `Org/Input.lean`: smooth representatives with the retained extraction.

These are the Osterwalder–Schrader forms: they pair observations with their reflected conjugates. For a matrix G, the notation $G \succeq 0$ means $\sum_{\alpha,\beta} \bar{c}_\alpha c_\beta G_{\alpha\beta} \geq 0$ for every complex vector c. Derivative and norm notation is that of Definition 8.12; m in the estimate below is a derivative order, independently of the gauge matrix size.

Proposition 8.14 (Euclidean properties)**.** *The kernels of Proposition 8.13 satisfy the following identities and bounds.*

(i) *For the coefficient action and units of Proposition 6.12, every permutation σ, $h \in H_4$, $b \in \mathbb{R}^4$ and distinct configuration x of (2.11),*

$$\begin{aligned} S_\varnothing &= 1, & S_{n+1}(u_d, v; x_0, x) &= S_n(v; x), \\ S_n(v \circ \sigma; x \circ \sigma) &= S_n(v; x), & S_n(\bar{v}; x) &= \overline{S_n(v; x)}, \\ S_n(\rho(h)v; hx + b) &= S_n(v; x). \end{aligned} \tag{8.29}$$

(ii) *For every coordinate direction and finite collection of positive configurations, the matrices in (8.27)–(8.28) satisfy*

$$\Big[G_S\big((v^\alpha, x^\alpha), (v^\beta, x^\beta)\big)\Big]_{\alpha,\beta} \succeq 0, \qquad [G_{\alpha\beta}(f)]_{\alpha,\beta} \succeq 0. \tag{8.30}$$

(iii) *For every $n, m \in \mathbb{N}$, degree list (d_i) and coefficient list $v_i \in V_{d_i}$, there are $C \geq 0$ and $b \in \mathbb{N}$ such that*

$$\|D_x^m S_n(v; x)\| \leq C r^{-b} \quad (0 < r \leq 1,\ \forall i \neq j,\ |x_i - x_j|_\infty \geq r)\,. \tag{8.31}$$

The identities descend through π_d to the quotient fields.

Proof. The proof checks the three clauses separately.

Step 1. Exact lattice identities pass to the limit at grid points; density and continuity extend them from coordinate-separated configurations to all of Conf_n.

Step 2. Lattice reflection positivity gives a positive semidefinite matrix for any finite list of joint positive-side tests. Each entry converges as a separated smear, so the limiting matrix is positive semidefinite; concentrating smooth tests gives point positivity.

Step 3. If $a_m < r/2$ and $6na_k < a_m$, the two first-crossing bounds give $a_k \geq c' r^4$. The centred estimate at scale a_k then yields a power bound in r, and derivative identities give the stated derivative bound.

For the first step, at grid configurations y and lattice translations b, the exact identities include

$$\begin{aligned} M_{N,p}(v \circ \sigma; y \circ \sigma) &= M_{N,p}(v; y), & M_{N,p}(\bar{v}; y) &= \overline{M_{N,p}(v; y)}, \\ M_{N,p}(\rho(h)v; hy + b) &= M_{N,p}(v; y) & & (h \in H_4,\ b \in \mathbb{Z}^4). \end{aligned}$$

The unit and empty-list identities pass in the same way. Equation (8.15) passes these formulas to the limit on Gen_n; density and smoothness extend them throughout Conf_n and to every real translation.

For the second step, fix finitely many joint tests supported in positive-side separated configurations and let $\mathcal{O}_{N,\alpha}$ denote their lattice smearings. Each insertion depends on finitely many links, whose physical distances from its insertion site tend to zero. Thus, for large p and then large N, these links lie strictly inside the positive half-torus, and Lemma 8.5 gives

$$\big[\langle \Theta_{r_\mu} \mathcal{O}_{N,\alpha}\, \mathcal{O}_{N,\beta} \rangle_{N,p}\big]_{\alpha,\beta} \succeq 0.$$

The two halves of each entry are internally separated and lie on opposite sides of the plane, so it is a separated smear. Pass each entry through the ordered limit (8.16); absolute convergence follows from smoothness on the compact supports, and the positive-semidefinite cone is closed. Therefore (8.30) holds for joint tests. Normalised smooth bumps concentrating at fixed positive configurations give the pointwise matrix assertion.

For the position-independent bound, (8.3) implies $a_{j+1} \geq c a_j^2$ for some $c > 0$, after adjusting finitely many levels. The first $j \geq j_0$ with $a_j < \delta \leq 1$ therefore satisfies $a_j \geq c_{j_0} \delta^2$. For $n \geq 1$ and fixed v, set $D = 1 + \max_i d_i$. Choose the first sufficiently large m with $a_m < \frac{r}{2}$, then the first $k \geq m$ with $a_k < \frac{a_m}{6(n+1)}$. The two first-crossing bounds give $a_k \geq c' r^4$. Equation (8.6) with $\varepsilon = 1$ and the separated-cell argument yield

$$|S(v;x)| \leq C r^{-4nD} \qquad (0 < r \leq 1,\ \forall i \neq j,\ |x_i - x_j|_\infty \geq r)\,.$$

First pass the estimate at points of (8.13), then use density, retaining separation $\frac{r}{2}$ before taking limits. Repeated derivative identities apply this bound to finitely many lists of higher-degree polynomials. A finite-dimensional norm comparison then bounds each Fréchet derivative norm. For no insertions use $S_\varnothing = 1$.[17] □

The null-space characterisation 8.13(ii) also gives the field symmetries. Inclusions commute with conjugation and H_4 and preserve unit classes. Derivatives commute with conjugation; the chain rule in $S(\rho(h)v; hx) = S(v;x)$ gives

$$\rho_{d+1}(h)\partial_\mu = \sum_\nu h_{\nu\mu}\partial_\nu \rho_d(h) \qquad (h \in H_4)\,.$$

Continuous rotation covariance will require a further argument.[18]

8.5 Removing the spacing refinement

Use the full fixed-volume convergence of Proposition 5.1. For a test supported inside volume p in Definition 2.11, write

$$\widehat{R}_p(v;f) = \lim_{N\to\infty} C_{N,p}(v;f)\,.$$

For every volume containing the test support, restriction to N_j and uniqueness in $\mathbb{C}$ give $\widehat{R}_p(v;f) = R_p(v;f)$, retaining the outer limit and smooth correlations. Start with the full spacing sequence and prescribed volume sequence p, and compose the volume refinements into one strictly increasing q. Then

$$\lim_{k\to\infty} \widehat{R}_{p(q(k))}(v;f) = \int f(x) S(v;x)\,dx\,. \tag{8.32}$$

At volumes not containing the support, assign arbitrary values as allowed by Definition 2.12; this gives the ordered Wilson limit for every coefficient list and separated test.[19]

Unresolved. The full-spacing fixed-volume theorem for the chosen Wilson family remains to be proved in the ultraviolet part, using its fixed-level Cauchy estimates and positional-limit construction. The uniqueness argument above uses this input; it does not prove it.[20]

[17] Formal correspondence at revision `3456e5e`: `Extraction/Harvest.lean`: symmetry and positivity; `Work/D05.lean`: two-scale bound; `Work/B10.lean`: derivative bounds.

[18] Formal correspondence at revision `3456e5e`: `Work/D06.lean`: orientation of the derivative maps.

[19] Formal correspondence at revision `3456e5e`: `FullVolume/Finale/Program.lean`: removal of the spacing refinement; `HierarchyBridge.lean`: public ordered Wilson limit.

[20] Formal correspondence at revision `3456e5e`: `FullVolume/Fields/Filled.lean`: full-spacing fixed-volume convergence for the chosen run.

The smooth null space agrees with the separated-test null space, so the public projection is $\pi(d, v) = (d, \pi_d v)$ by Definition 2.17. This identification in separated correlations transports (8.32), after multiplication by f, to the field correlations of Proposition 5.2; its phase clauses use the later construction on this same table.

9 Free Maxwell fields

The limiting fields of Section 8 are described by their correlations. To identify their behaviour at short distances, we compare them with a model whose correlations are explicit: the free Maxwell field, with one component for each Lie algebra direction in Definition 2.1. Its local observables are polynomials in the curvature and its derivatives, subject to the linear Maxwell equations.

We first construct those polynomial correlations in Subsection 9.1. Subsection 9.2 identifies their null coefficients and explains why finitely many smeared observations detect every field of a fixed degree. Subsection 9.3 states the precise lattice comparison estimate needed from the ultraviolet construction and passes it to continuum fields. Subsection 9.4 turns the resulting maps into compatible isomorphisms. Subsection 9.5 then recovers convergence of all positional derivatives. The radii and field maps are chosen together; a prescribed sequence of radii requires further estimates used later in the trace argument.

The polynomial positivity argument uses the estimate for reflected linear observations in Lemma 9.6. The comparison with the interacting fields uses the lattice estimates (9.17)–(9.18) and the separate dimension bound (9.19). Given these estimates, finite-dimensional linear algebra identifies the field spaces and their short-distance correlations.

9.1 Curvature correlations

Use the gauge group of Definition 2.1, the classical polynomial spaces V_d of Definition 6.4, and their norms from Definition 8.1. The physical inner product and a corresponding colour basis $E^1, \dots, E^{\dim \mathfrak{g}}$ are normalised by

$$\langle X, Y\rangle_{\mathfrak{g}} = -\frac{1}{m}\operatorname{Re}\operatorname{tr}(XY), \qquad \langle E^c, E^{c'}\rangle_{\mathfrak{g}} = \delta_{cc'}. \tag{9.1}$$

Here m is the defining matrix size, as in Definition 2.1; the basis component c is called a colour index.[21]

Definition 9.1 (Maxwell Taylor coefficients)**.** The indices a, b, k range over the four coordinate directions. A Maxwell field is a smooth array $f = (f_{ab})$ of $\mathfrak{g}$-valued functions, viewed as a two-form by antisymmetry, satisfying

$$f_{ab} = -f_{ba}, \qquad \partial_a f_{bk} + \partial_b f_{ka} + \partial_k f_{ab} = 0, \qquad \sum_a \partial_a f_{ab} = 0. \tag{9.2}$$

For $j \in \mathbb{N}$, let $\mathcal{H}_j$ be the finite-dimensional real space of homogeneous polynomial solutions of degree j. A homogeneous polynomial Q_j is identified with its symmetric multilinear coefficient by evaluation on repeated arguments. Our Taylor convention is

$$Q_j(z) = \frac{1}{j!} D^j f(0)[z, \dots, z].$$

[21] Formal correspondence at revision `3456e5e`: `YM/Gauge/Colors.lean`, lines 43–83: physical colour coordinates and normalised trace.

Thus a Taylor tensor here includes the factorial denominator. A sequence $Q = (Q_j)_{j\ge 0}$ with $Q_j \in \mathcal{H}_j$ is a formal Maxwell Taylor series; convergence of this series is not required. For a one-form A, use $z \cdot A = \sum_a z^a A_a$ and $(dA)_{ab} = \partial_a A_b - \partial_b A_a$. The radial potential associated with $Q \in \mathcal{H}_j$ is

$$(\kappa_j Q)_a(z) = \frac{1}{j+2} \sum_b z^b Q_{ba}(z), \qquad z \cdot \kappa_j Q(z) = 0, \qquad d(\kappa_j Q) = Q. \tag{9.3}$$

The last identity follows by differentiating the first formula, using the cyclic equation in (9.2) and Euler's identity for a homogeneous polynomial. Thus the coefficients may also be read in the radial coordinates of (6.11).[22]

For $z \in \mathbb{R}^4 \setminus \{0\}$, the scalar Green function determines the covariance, or two-point pairing, between curvature components at distinct points. The indices a, b, e, f specify coordinate directions; c, c' are the colour indices of (9.1). Set

$$\begin{aligned} \Gamma(z) &= \frac{1}{4\pi^2 |z|^2}, \qquad z \neq 0, \\ C^{cc'}_{ab,ef}(z) &= \delta_{cc'}\big(\delta_{be}\partial_a\partial_f\Gamma - \delta_{bf}\partial_a\partial_e\Gamma - \delta_{ae}\partial_b\partial_f\Gamma + \delta_{af}\partial_b\partial_e\Gamma\big)(z). \end{aligned} \tag{9.4}$$

The signs include differentiation at the second endpoint of a covariance depending on $x - y$. A linear probe of order j is a real linear functional on $\mathcal{H}_j$: it measures a specified linear combination of the Taylor coefficients. For probes $p \in \mathcal{H}_j^*$ and $q \in \mathcal{H}_\ell^*$, the dual spaces of these real tensor spaces, define

$$K_{p,q}(z) = \frac{(-1)^\ell}{j!\,\ell!}(p \otimes q)\big[D^{j+\ell} C(z)\big]. \tag{9.5}$$

The first j derivative slots belong to p and the remaining ℓ slots to q. Colour and form indices are contracted by the two functionals themselves; no extra antisymmetry factor is inserted. One may extend these functionals to unconstrained tensors to evaluate the formula. The following observation shows that the answer is independent of those extensions.

Lemma 9.2 (The contraction kernel)**.** *For $z \neq 0$, the kernel (9.4) satisfies (9.2) in both groups of tensor indices. Consequently (9.5) is well defined on $\mathcal{H}_j^* \times \mathcal{H}_\ell^*$, is real analytic, and satisfies*

$$K_{p,q}(z) = K_{q,p}(-z), \qquad K_{p,q}(\lambda z) = \lambda^{-j-\ell-4} K_{p,q}(z) \quad (\lambda > 0). \tag{9.6}$$

Proof. The verification has three steps.

Step 1. Taking a curl at both endpoints gives the displayed covariance; antisymmetry and commuting derivatives give the cyclic equation.

Step 2. Harmonicity of Γ gives the divergence equation.

Step 3. Endpoint exchange and homogeneity give (9.6); the equations make the contraction independent of how each functional is extended.

For the second step, write $\Delta = \sum_a \partial_a^2$ for the Laplacian and suppress the colour factor:

$$\sum_a \partial_a C_{ab,ef} = \delta_{be}\partial_f \Delta\Gamma - \delta_{bf}\partial_e \Delta\Gamma = 0 \qquad (z \neq 0).$$

[22] Formal correspondence at revision `3456e5e`: `YM/Maxwell/Shell.lean`: the constrained Taylor space and inverse radial curl.

Antisymmetry is explicit in (9.4). Exchanging the endpoints proves the same equations in the other group of indices. Every Taylor tensor to which an extended functional is applied therefore lies in $\mathcal{H}_j$ or $\mathcal{H}_\ell$. An extension vanishing on that space contributes zero, proving independence. The exchange and scaling identities follow by differentiating the even, homogeneous function Γ; analyticity follows from its explicit formula.[23] □

Definition 9.3 (Free polynomial fields)**.** Choose coordinates on the spaces $\mathcal{H}_j$ of Definition 9.1 and give each coordinate weight $j+2$. Substitution of (9.3) in a radial polynomial defines a linear map res_d on V_d. Write

$$\begin{aligned}(\operatorname{res}_d v)(Q) &= v\big((\kappa_j Q_j)_{j\ge 0}\big),\\ \mathcal{I}_d &= \ker \operatorname{res}_d, \qquad \mathcal{P}_d^0 = \operatorname{res}_d(V_d), \qquad F_d^0 = V_d/\mathcal{I}_d \simeq \mathcal{P}_d^0.\end{aligned} \tag{9.7}$$

For the orthogonal group $O(4)$ of Definition 6.5, write $\rho_d^0(h)$ for the action induced on these free polynomial spaces by pullback of the two-form and its Taylor tensors. Give each F_d^0 its fixed quotient norm from V_d; all vector and operator norms below use this norm on the free factors. The free unit is the constant class $\mathbf{1} = [1]$. The interacting unit is $\pi_d u_d$ in Definition 8.11; the comparison maps below will send the free unit to that class.

Wick ordering removes pairings between coefficients at the same insertion point. Only pairings between distinct observations remain in their correlation. The following finite sum implements this rule.

Definition 9.4 (Wick correlations)**.** For monomials $q_i = \prod_{e\in E_i} p_e$ in the linear probes defined above, let E be their disjoint union and let $i(e) = i$ identify the insertion containing e. A perfect matching is a partition of E into pairs. Let $\mathfrak{M}_{\neq}(E)$ be the matchings that never pair two members of the same E_i. Set

$$W_n(q;x) = \sum_{P\in\mathfrak{M}_{\neq}(E)} \prod_{\{e,f\}\in P} K_{p_e,p_f}\big(x_{i(e)} - x_{i(f)}\big), \qquad x \in \operatorname{Conf}_n, \tag{9.8}$$

For $q = (q_1, \dots, q_n)$, extend this formula complex multilinearly to the polynomial fields of Definition 9.3.

The configuration space is that of Definition 2.10. The matching rule treats repeated occurrences separately and is unchanged by their enumeration. For example, a quadratic insertion has zero one-point expectation, whereas two quadratic insertions give

$$W_2(pq, st; x, y) = K_{p,s}(x-y)K_{q,t}(x-y) + K_{p,t}(x-y)K_{q,s}(x-y).$$

No value of a kernel at zero is used. Empty products give $W_0 = 1$.[24]

9.2 Positivity and finite observations

Reflection positivity must hold for polynomials, not just for the linear curvature. The analytic input concerns linear probes; finite-dimensional Gaussian algebra will then handle every polynomial.

[23] Formal correspondence at revision `3456e5e`: `YM/Maxwell/Curls.lean`: equations in both groups of tensor indices; `YM/Maxwell/Leg.lean`, lines 141–176: independence of the extensions.

[24] Formal correspondence at revision `3456e5e`: `YM/Maxwell/Equation.lean`: restriction and equation kernel; `YM/Maxwell/Wick.lean`: matching correlations.

Definition 9.5 (Reflected Maxwell observations)**.** Fix an axis μ and let $\theta = r_{\mu,0}$ be the reflection of Definition 8.2. A positive-side point has $x^\mu > 0$. Reflect the two-form indices and all Taylor arguments by

$$(\theta \cdot Q)_{ab}(z) = \sum_{c,d=1}^{4} \theta_{ac}\theta_{bd} Q_{cd}(\theta^{-1} z), \qquad (R_\theta p)(Q) = p(\theta^{-1} \cdot Q).$$

The colour component is unchanged. Extend R_θ multiplicatively to polynomials; on F_d^0 it is $\rho_d^0(\theta)$ of Definition 9.3. For a finite list of probes p_α at positive-side points x_α, their reflected Gram matrix, the matrix of reflected pairings, is

$$H_{\alpha\beta} = K_{p_\alpha, R_\theta p_\beta}\big(x_\alpha - \theta x_\beta\big). \tag{9.9}$$

The reflected point is separated from every positive-side point, even when two of the x_α coincide.

Lemma 9.6 (Reflected linear probes)**.** *The matrix* (9.9) *is positive semidefinite. If all $x_\alpha = x$ and the probes p_α are linearly independent as functionals on $\prod_j \mathcal{H}_j$, it is positive definite.*

Unresolved. The analytic proof of Lemma 9.6 is still missing from this manuscript. It must prove nonnegativity for arbitrary positive-side probe lists and strictness at one point modulo exactly the Maxwell equations. The source uses a nonnegative energy integral, analytic continuation of a vanishing response and tests against homogeneous Maxwell solutions; positivity alone does not establish the strict assertion.[25]

The strict assertion detects every nonzero local polynomial, including its constant term. It therefore identifies the quotient (9.7) with the quotient by all separated null correlations, just as required for the interacting fields in 8.13(ii).

The inclusions and derivatives below are defined directly on the polynomial realisation $\mathcal{P}_d^0$ of Definition 9.3.

Definition 9.7 (Free field operations)**.** Write $j_{d,e}^0$ for the inclusion induced by $V_d \subset V_e$, for $d \le e$. For a formal Taylor series Q as in Definition 9.1, its infinitesimal translation in direction μ has components

$$(\mathcal{D}_\mu Q)_j = (j+1) Q_{j+1}(e_\mu, \cdot, \dots, \cdot).$$

Here e_μ is the coordinate unit vector and the remaining arguments are the j symmetric Taylor arguments; form indices are retained. For a polynomial q, let $D_j q$ denote its derivative in the Q_j variable and define

$$(\partial_\mu^0 q)(Q) = \sum_j D_j q(Q)\big[(\mathcal{D}_\mu Q)_j\big]. \tag{9.10}$$

The sum is finite, since q uses only finitely many Q_j.

The proof below shows that this derivative maps $\mathcal{P}_d^0$ into $\mathcal{P}_{d+1}^0$ and gives the positional derivative of its correlations. In derivative bounds, D_x^s denotes the total Fréchet derivative with its operator norm, as in Definition 8.12.

Proposition 9.8 (Free fields)**.** *Grant Lemma 9.6. The correlations* (9.8) *descend faithfully to F_d^0 in* (9.7) *and have the following properties.*

[25] Formal correspondence at revision `3456e5e`: `YM/Next/MAXWELL/Gram.lean`, lines 52–108; `YM/Next/MAXWELL/Strict.lean`, especially `level_detects`, and its imported energy estimates.

(i) *They are normalised, unital and real analytic on* Conf_n*; they satisfy the symmetry identities* (8.29)*, with the full* $O(4)$ *action on coefficients, and both reflection-positivity conditions* (8.30)*.*

(ii) *The operations of Definition 9.7 give compatible inclusions* $j^0_{d,e} : F^0_d \hookrightarrow F^0_e$ *and derivatives* $\partial^0_\mu : F^0_d \to F^0_{d+1}$ *satisfying* (8.26) *and the conjugation and orientation laws of Section 8.4.*

(iii) *For every fixed field list and* $s \in \mathbb{N}$*, there are* $C \geq 0$ *and* $b \in \mathbb{N}$ *with*

$$\|D^s_x W_n(q;x)\| \leq C r^{-b} \quad \big(0 < r \leq 1,\ |x_i - x_j|_\infty \geq r \text{ for } i \neq j\big). \tag{9.11}$$

Proof. The proof has four steps.

Step 1. Factor the real reflected matrix as $H = L^{\mathsf{T}} L$; the matching rule then gives a polynomial inner product.

Step 2. Strictness at one point makes this representation injective, so the only null coefficients are $\mathcal{I}_d$.

Step 3. The same finite matching formula gives positivity, normalization and symmetries.

Step 4. Differentiating the matching formula gives the field operations and the homogeneous-kernel bounds.

For a finite collection of positive-side products, use the finite set of all their linear occurrences as the matrix indices. Within each product, sum over partial matchings between distinct insertion points, and replace each unpaired occurrence e by $\sum_a L_{ae} z_a$. This produces a polynomial $Q_\alpha(z)$ for each product α. Every full matching splits uniquely into a matching within each half and the pairs crossing between them. Write $P(z) = \sum_\gamma P_\gamma z^\gamma$, where γ is a multi-index of nonnegative integers and $z^\gamma = \prod_a z_a^{\gamma_a}$. The reflected Gram matrix is then

$$G_{\alpha\beta} = \langle Q_\alpha, Q_\beta \rangle_{\mathrm{F}}, \qquad \langle P, Q \rangle_{\mathrm{F}} = \sum_\gamma \gamma!\, \overline{P_\gamma} Q_\gamma, \qquad \gamma! = \prod_a \gamma_a!. \tag{9.12}$$

This proves point positivity, since every weight $\gamma!$ is positive. For a single insertion there are no internal pairs, so $Q(z) = q(L^{\mathsf{T}} z)$. For independent coordinate probes at one point, Lemma 9.6 makes L invertible and hence

$$q \neq 0 \quad \Longrightarrow \quad W_2\big(R_\theta \bar{q}, q; \theta x, x\big) = \sum_\gamma \gamma!\, |Q_\gamma|^2 > 0. \tag{9.13}$$

Thus the only separated null coefficients in V_d are $\mathcal{I}_d$. Integrating point Gram matrices against finite sums of positive-side joint tests gives smeared positivity. The integrals converge absolutely on the separated compact supports, by the explicit kernels.

The constant polynomial and the matching rule give the normalisation and unit identities. The kernel is invariant under translations and transforms as a two-form under $O(4)$; permuting occurrences proves permutation symmetry and conjugating coefficients proves reality. All these identities descend to the quotient.

For differentiation, use the infinitesimal translation of Definition 9.7. It raises weight by one, obeys the Leibniz rule, and preserves (9.2). The induced derivation preserves the images $\mathcal{P}^0_d$, and differentiating a matching sum in slot i gives

$$\partial_{i,\mu} W_n(q;x) = W_n(q_{i \to \partial^0_\mu q_i};x).$$

Inclusions are induced by $V_d \subset V_e$; faithfulness makes them injective and compatible. Finally, each derivative of a contraction is homogeneous of a fixed negative degree and bounded on the unit sphere. The finite sums of products in (9.8) give (9.11), including the empty-list case.[26] □

Faithfulness can be tested with finitely many observations at each degree. This converts convergence of correlations into information about coefficient maps, by using only finitely many measurements.

Definition 9.9 (Tested observations)**.** For $v \in F_d^0$, a fixed list $b = (b_1, \dots, b_n)$ of free fields in the other insertion slots, and a separated test f of Definition 2.10, write

$$\ell(v) = \int f(x) W_{n+1}(v, b; x)\, dx. \tag{9.14}$$

Here (v, b) is the list obtained by placing v first.

Lemma 9.10 (Finite separating tests)**.** *Assume Lemma 9.6. For each d, finitely many functionals (9.14) and a constant $c_d > 0$ satisfy*

$$\max_{1 \le a \le s_d} |\ell_a(v)| \ge c_d \|v\| \qquad (v \in F_d^0). \tag{9.15}$$

Proof. The proof has two steps.

Step 1. By (9.13), every nonzero field has a correlation detected by some test, so the common kernel of all functionals (9.14) is zero.

Step 2. Choose each next functional nonzero on the current common kernel; its dimension drops at every step, leaving finitely many tests, and compactness of the unit sphere gives the positive constant in (9.15).

In detail, starting with F_d^0, choose a functional nonzero on the current common kernel until its dimension reaches zero:

$$E_0 = F_d^0, \qquad E_{a+1} = E_a \cap \ker \ell_{a+1}, \qquad E_a \neq 0 \implies \dim E_{a+1} < \dim E_a.$$

Only finitely many choices are needed. The maximum in (9.15) is then positive on the unit sphere; its minimum there is $c_d > 0$.[27] □

9.3 The comparison estimate

Return to the same limiting correlations S of Proposition 8.13 and the normed interacting fields F_d of Definition 8.11. A map from a free field to an interacting field is useful only if it compares their correlations. The needed estimate is uniform in the size of a joint test, including tests at distinct points with equal coordinate projections. For $R \ge 1$, put

$$K_{R,n} = \{x : |x| \le R,\ |x_i - x_j|_\infty \ge R^{-1} \text{ for } i \neq j\}. \tag{9.16}$$

Every compact subset of Conf_n lies inside such a set for sufficiently large R.

[26] Formal correspondence at revision `3456e5e`: `YM/Next/MAXWELL/Fock.lean`, `YM/Next/MAXWELL/Studio.lean` and `YM/Next/MAXWELL/Core.lean`: polynomial factorisation and strictness; `YM/Next/MAXWELL/System.lean`, lines 29–114: quotient and derivative operations; `YM/Next/MAXWELL/Caps.lean`: analyticity and bounds.

[27] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Invert.lean`, lines 48–65; `YM/Work/A04.lean`, lines 14–36.

Definition 9.11 (Finite-cutoff comparison)**.** The comparison data at finite lattice spacing have the following quantifiers. For every $D, N_*, R, \varepsilon, \delta$ with $D, N_* \in \mathbb{N}$, $R \geq 1$ and $\varepsilon, \delta > 0$, require a radius $0 < r \leq \delta$ and linear maps

$$A_{N,p,d} : F_d^0 \longrightarrow V_d \qquad (d \leq D). \tag{9.17}$$

These coefficient maps choose lattice polynomials representing free fields. Their operator norms are eventually bounded on the ordered tails of Definition 8.3; they commute with conjugation and signed permutations and preserve the degree-zero unit there. For tuples with specified degrees, abbreviate $A_{N,p}q = (A_{N,p,d_i}q_i)_i$. For a point list x, write $rx = (rx_1, \dots, rx_n)$. For every $n \leq N_*$, $d_i \leq D$, fixed tuple $q_i \in F_{d_i}^0$, separated test f supported in $K_{R,n}$, $M > 0$ with $\|f\|_\infty \leq M$, and $\eta > 0$, require eventually

$$\left| C_{N,p}\big(A_{N,p}q; f_r\big) - \int f(x) W_n(q; x)\, dx \right| \leq \varepsilon M \prod_i \|q_i\| + \eta, \qquad f_r(y) = r^{-4n} f\Big(\frac{y}{r}\Big). \tag{9.18}$$

The moments and tests are those of Definitions 2.9 and 2.10. The radius and maps precede the choice of tuple and test. The cutoff threshold may depend on that tuple, test, M and η; no common cutoff threshold for all tests is required.

We also need the independent dimension bound

$$\dim_{\mathbb{C}} F_d \leq \dim_{\mathbb{C}} F_d^0 \qquad (d \in \mathbb{N}). \tag{9.19}$$

Unresolved. The ultraviolet sections must produce (9.17)–(9.18) and (9.19) for the same Wilson family and the same retained limits as in Section 8. The comparison includes bounded coefficient maps, symmetries, the degree-zero unit and all the displayed test quantifiers. Centred bounds alone do not provide it.[28]

For maps $U_d : F_d^0 \to F_d$, write $Uq = (U_{d_i} q_i)_i$, just as for the coefficient maps in Definition 9.11. First pass that comparison to the already constructed continuum theory. Bounded coefficient maps have a common cluster limit in finite dimensions. The extraction is used to estimate the same S, rather than to construct another limiting law.

Lemma 9.12 (Passage to continuum fields)**.** *Assume* (9.17)–(9.18) *and the conclusions of Propositions 8.8, 8.13 and 8.14. There are linear maps $U_d : F_d^0 \to F_d$, $d \leq D$, commuting with conjugation and signed permutations, with $U_0\mathbf{1} = \mathbf{1}$, such that*

$$\left| \int f(x) \big[S_n(Uq; rx) - W_n(q; x) \big]\, dx \right| \leq \varepsilon M \prod_i \|q_i\| \tag{9.20}$$

for all the tuples and tests in (9.18)*.*

Proof. There are two steps.

Step 1. The bounded finite-dimensional maps have a common cluster limit A_d on the ordered tails; set $U_d = \pi_d A_d$.

[28] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Station.lean`: exact finite-cutoff comparison condition; `YM/Next/UV_STRESS/Granular.lean` and `YM/MergeProof/East/Plume.lean`: suppliers; `YM/Next/UV_STRESS/Invert.lean`, lines 104–116: the separate dimension input.

Step 2. Local bounds replace the varying coefficients by A_d in correlations, and the ordered limit turns the finite-cutoff estimate into (9.20) for the same S.

More precisely, use a refining ultrafilter as in Definition 8.3 and write its limits as A_d. Expansion in fixed polynomial bases and (8.17) give, for the fixed rescaled test,

$$|C_{N,p}(A_{N,p}q; f_r) - C_{N,p}(Aq; f_r)| \le C_{r,f,q} \sum_{d\le D} \|A_{N,p,d} - A_d\| \longrightarrow 0.$$

Put $U_d = \pi_d A_d$, using the quotient map of Proposition 8.13. The retained ordered limit and the change of variables $y = rx$ yield

$$C_{N,p}(A_{N,p}q; f_r) \longrightarrow \int f(x) S_n(Uq; rx)\, dx.$$

All symmetry and unit identities pass through the finite-dimensional limits. For each fixed test, take the limit in (9.18), then let $\eta \downarrow 0$. The same cluster maps work for every test because they were chosen before any test was fixed.[29] □

9.4 Compatible field bases

We next choose comparisons at successively smaller radii. The maps obtained in Lemma 9.12 need not initially agree on a field that is included in two different degrees. We correct this discrepancy on the lower-degree subspaces before proving invertibility.

For a radius sequence r_k and maps $U_{k,d} : F_d^0 \to F_d$, write

$$E_k(q; x) = S_n(U_k q; r_k x) - W_n(q; x). \tag{9.21}$$

Definition 9.13 (Uniform tested convergence)**.** For every fixed field tuple, the convergence used below is

$$\sup_{\substack{f \text{ as in Definition 2.10} \\ \operatorname{supp} f \subset K,\ \|f\|_\infty \le 1}} \left| \int f(x) E_k(q; x)\, dx \right| \longrightarrow 0 \qquad (K \Subset \mathrm{Conf}_n), \tag{9.22}$$

Here $K \Subset \mathrm{Conf}_n$ means a compact subset of the separated configuration space.

The supremum controls all tests with the same size bound on K, so this is stronger than convergence for each test separately.

Definition 9.14 (Moving tested observations)**.** For a finite list of tested observations ℓ_a on F_d^0 from Definition 9.9, let f_a be its test and b_a its list of m_a other fields. Given r_k and maps $U_{k,d}$, define

$$\ell_{a,k}(v) = \int f_a(x) S_{m_a+1}\big(U_k(v, b_a); r_k x\big)\, dx, \qquad B_k v = (\ell_{a,k}(v))_a, \quad Bv = (\ell_a(v))_a. \tag{9.23}$$

These are the same measurements, now made on the interacting fields at radius r_k.

Proposition 9.15 (Compatible comparisons)**.** *Assume the finite-cutoff input* (9.17)–(9.18)*, the dimension bound* (9.19)*, and the field operations and Euclidean properties of Propositions 8.8, 8.13, 8.14 and 9.8. There exist $r_k \downarrow 0$ strictly and complex-linear isomorphisms $U_{k,d} : F_d^0 \to F_d$ satisfying* (9.22)*. For each fixed $d \le e$, eventually in k,*

$$\begin{aligned} U_{k,e} j^0_{d,e} &= j_{d,e} U_{k,d}, & U_{k,d}\mathbf{1} &= \mathbf{1}, \\ U_{k,d}\bar{q} &= \overline{U_{k,d} q}, & U_{k,d}\rho_d^0(h) &= \rho_d(h) U_{k,d} \quad (h \in H_4). \end{aligned} \tag{9.24}$$

[29] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Station.lean`, especially `zoom_integral` and `harvested`.

Proof. The construction has four steps.

Step 1. At stage k, require accuracy through degree and number of insertions k, and choose $r_k \downarrow 0$.

Step 2. Correct each map on the lower-degree image using a left inverse, so $\widetilde{U}_{k,d+1} j_d^0 = j_d \widetilde{U}_{k,d}$ while preserving conjugation and signed symmetries.

Step 3. Finite separating tests give $\|B_k v\|_\infty \geq (c_d/2)\|v\|$, hence each fixed-degree map is eventually injective.

Step 4. The independent dimension bound gives $\dim F_d^0 \leq \dim F_d \leq \dim F_d^0$, hence eventual isomorphisms degree by degree.

Apply Lemma 9.12 with

$$D_k = N_{*,k} = k, \qquad R_k = k+1, \qquad \varepsilon_k = \frac{1}{k+1}, \qquad r_{k+1} \leq \min\Big(\frac{r_k}{2}, \frac{1}{k+2}\Big).$$

The comparison hypothesis permits the last condition by choosing the next radius bound first. Each compact separated set eventually fits inside $K_{R_k,n}$. For each fixed tuple, (9.20) therefore gives (9.22) for the initial maps.

Write $j_d^0 = j_{d,d+1}^0$ and $j_d = j_{d,d+1}$. Choose a linear left inverse $R_d : F_{d+1}^0 \to F_d^0$, so $R_d j_d^0 = 1$. It extracts the part of a higher-degree field already present at degree d. It may be chosen real and commuting with the H_4 action of Definition 8.1: average any left inverse first with its conjugate and then over H_4. Both averages remain left inverses because j_d^0 intertwines these symmetries. For $d < D_k$, define corrected maps recursively by

$$\widetilde{U}_{k,0} = U_{k,0}, \qquad \widetilde{U}_{k,d+1} = j_d \widetilde{U}_{k,d} R_d + U_{k,d+1}(1 - j_d^0 R_d). \tag{9.25}$$

Since $R_d j_d^0 = 1$, this gives $\widetilde{U}_{k,d+1} j_d^0 = j_d \widetilde{U}_{k,d}$. The symmetry identities pass through the recursion; the degree-zero unit identity propagates to every degree. For (9.22), expand a corrected correlation using

$$q = j_d^0 R_d q + (q - j_d^0 R_d q).$$

After finitely many expansions it is a sum of original comparisons with fixed free fields, possibly in lower degrees. The inclusion identities in both theories identify the corresponding sum of free correlations with $W_n(q; x)$. Thus the corrected maps retain (9.22).

Fix d and choose the finite functionals of Lemma 9.10. Use their maps $B_k : F_d^0 \to \mathbb{C}^{s_d}$ and B from Definition 9.14, with the corrected comparisons $\widetilde{U}_k$. Then (9.22), evaluated on a basis, gives operator-norm convergence to $B = (\ell_a)_a$. For large k, (9.15) implies

$$\|B_k v\|_\infty \geq \frac{c_d}{2}\|v\|, \qquad \widetilde{U}_{k,d} v = 0 \implies B_k v = 0 \implies v = 0.$$

Hence $\widetilde{U}_{k,d}$ is injective. Together with (9.19),

$$\dim F_d^0 \leq \dim F_d \leq \dim F_d^0,$$

so it is an isomorphism eventually at each fixed degree. Fill in or replace the finitely many exceptional early terms at each degree by any fixed isomorphism. For a fixed tuple or a fixed pair of degrees these replacements have no effect on the eventual assertions, proving (9.24) and (9.22).[30] □

[30] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Rates.lean`: diagonal choice of radii; `YM/Next/UV_STRESS/Arrange.lean`: equivariant retractions and exact degree correction; `YM/Next/UV_STRESS/Invert.lean`, especially `Stations.equivalences`: eventual injectivity and isomorphisms.

9.5 Derivatives of the comparison

The tested convergence (9.22) also determines how the derivatives of the comparison fields change with k. Integration by parts identifies their values against every separating functional; finite-dimensional invertibility then identifies the derivative maps themselves. This is the extra argument needed before using a smooth short-distance limit.

Proposition 9.16 (Smooth short-distance limits)**.** *Under the hypotheses of Proposition 9.15, for every $d \in \mathbb{N}$ and coordinate direction μ, the maps there satisfy, in operator norm,*

$$r_k U_{k,d+1}^{-1} \partial_\mu U_{k,d} \longrightarrow \partial_\mu^0. \tag{9.26}$$

For every fixed tuple q, compact $K \subset \mathrm{Conf}_n$ and $s \in \mathbb{N}$,

$$\sup_{x \in K} \|D_x^s E_k(q;x)\| \longrightarrow 0. \tag{9.27}$$

Proof. The proof has three steps.

Step 1. Integration by parts identifies the transported derivative through the finite tests: $B_k D_{k,d,\mu} q \to B\partial_\mu^0 q$.

Step 2. Repeated differentiation and finite-dimensional multilinearity give uniform tested convergence for every derivative list, hence $\|D^\alpha E_k\|_{L^1(K)} \to 0$.

Step 3. A compactly supported cutoff and Fourier inversion give $\|D^s(\chi E_k)\|_\infty \to 0$ on the target compact set.

Put $D_{k,d,\mu} = r_k U_{k,d+1}^{-1} \partial_\mu U_{k,d}$. The field derivative identity (8.26) and the chain rule give

$$\partial_{i,\mu} S_n(U_k q; r_k x) = S_n\big(U_k(q_{i \to D_{k,d_i,\mu} q_i}); r_k x\big).$$

Use the finite tested observations at degree $d+1$ from Lemma 9.10, with $\ell_{a,k}, f_a, b_a, m_a$ as in Definition 9.14. Integration by parts in the first insertion variable gives

$$\ell_{a,k}(D_{k,d,\mu} q) = -\int \partial_{1,\mu} f_a(x) S_{m_a+1}\big(U_k(q, b_a); r_k x\big)\, dx \longrightarrow \ell_a(\partial_\mu^0 q). \tag{9.28}$$

The maps B_k of Definition 9.14 converge in operator norm to the injective map $B = (\ell_a)_a$, as in the preceding proof. Choose a left inverse L of B, so $LB = 1$. Then $LB_k \to 1$, and for large k,

$$D_{k,d,\mu} q = (LB_k)^{-1} L\big(B_k D_{k,d,\mu} q\big) \longrightarrow \partial_\mu^0 q.$$

Convergence on a finite basis gives (9.26).

Repeated differentiation expresses each positional derivative in terms of higher-degree fields whose free coordinates converge, by (9.26). Multilinearity and expansion in finite bases extend (9.22) to these converging coordinate lists. For a coordinate multi-index α and compact $K \Subset \mathrm{Conf}_n$, this gives

$$\sup_{\substack{f \text{ as in Definition 2.10} \\ \operatorname{supp} f \subset K,\ \|f\|_\infty \le 1}} \left| \int f(x) D^\alpha E_k(q;x)\, dx \right| \longrightarrow 0.$$

Take a slightly larger compact neighbourhood of K inside Conf_n. Approximating bounded measurable tests there by smooth tests gives the dual formula for the integral of an absolute value, hence

$$\|D^\alpha E_k(q;\cdot)\|_{L^1(K)} \longrightarrow 0.$$

Choose a smooth cutoff function χ supported in that larger set and equal to one near K. Writing $a = 4n$ for the dimension of the point configuration space and α for a coordinate multi-index, integration by parts in the Fourier transform and Fourier inversion give

$$\|D^s(\chi E_k)\|_\infty \le C_{\chi,s} \sum_{|\alpha|\le s+a+1} \|D^\alpha(\chi E_k)\|_{L^1} \longrightarrow 0.$$

This proves (9.27); the zero-dimensional case is just convergence of constants.[31] □

For simple nonabelian $\mathfrak{g}$, no gauge-invariant linear functional on $\mathfrak{g}$ is nonzero. Since a nonconstant polynomial of weight below four could only be linear, this determines the lowest field spaces.

Corollary 9.17 (Degrees below four)**.** *If $\mathfrak{g}$ is simple in Definition 2.1, then, for $d < 4$,*

$$F_d^0 = \mathbb{C}\mathbf{1}, \qquad \dim_{\mathbb{C}} F_d^0 = 1. \tag{9.29}$$

Under (9.19), the interacting space also satisfies $F_d = \mathbb{C}\mathbf{1}$.

Proof. There are two steps.

Step 1. Below weight four, only a single curvature or first-derivative coefficient could occur; gauge invariance would require a linear functional $\lambda : \mathfrak{g} \to \mathbb{C}$.

Step 2. Since $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, such a functional vanishes, leaving only constants; the unit survives in both quotients under the stated dimension bound.

Indeed, simplicity and noncommutativity imply $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, so

$$\lambda([X, Y]) = 0 \text{ for all } X, Y \in \mathfrak{g} \quad \Longrightarrow \quad \lambda = 0.$$

Thus V_d consists of constants, and $W_1(1) = 1$ shows that the constant survives the free quotient. The dimension bound and the nonzero interacting unit from (8.29) give the last assertion.[32] □

The construction of Proposition 9.15 chooses its radii to satisfy successive finite requirements. It does not ensure that those radii form a subsequence of a previously specified sequence. That stronger choice requires comparison estimates uniform over varying scale intervals, and is reserved for the trace section. Identifying and normalising the degree-four observable of Proposition 5.5 requires further identities for differentiated correlations. Those later arguments use the field spaces and derivative limits established here.

References

[1] OpenAI, *Lean construction of four-dimensional Yang–Mills theory*, Source release, revision `3456e5e`, 2026, https://github.com/yang-mills-group/OpenAI-YM-20261028.

[31] Formal correspondence at revision `3456e5e`: `YM/Next/UV_STRESS/Focus.lean`, especially `head_tilt`, `bend_limit` and `travels_jet`; `YM/Work/A04.lean`, `tendsto_of_fitted`: recovery from varying finite tests.

[32] Formal correspondence at revision `3456e5e`: `YM/UVNext/FREE_STRESS_CERTIFICATE/Size.lean`, `just_one` and `dim_before`; `YM/Trace2/Transfer.lean`, `first_grade`.